# What the data pulls show about Aladdin, Aladdin Studio and BlackRock

Prepared October 19, 2026. Independent research compiled from BlackRock's public material. Not produced, reviewed or endorsed by BlackRock.

## What this document is

A plain statement of what was learned about BlackRock's Aladdin platform, Aladdin Studio and BlackRock itself from the material pulled for this project. It is organised in that order. It lays out facts and the source of each fact, and it says where the material is silent. It gives no advice.

## What was pulled

| Material | What it is | Size |
|---|---|---|
| API specifications | The 54 Swagger 2.0 specifications inside BlackRock's 11 public `asdk_plugin_*` packages on PyPI (version 2.0.0b3), plus the AladdinSDK README | 54 specs, 301 operations |
| Product pages | BlackRock's Aladdin, Aladdin Studio, Studio for Risk, Copilot, Whole Portfolio, Wealth, eFront and APIs pages; the Fivetran, Microsoft and Snowflake write-ups | 15 pages |
| Transcripts, part 1 | The requested YouTube videos and BlackRock site videos: the Snowflake Summit talk, the LangChain Interrupt talk, Never Done podcast episodes with Microsoft and on Studio and Snowflake, the Big Little Questions series, Aladdin Explore and eFront explainers | 33 files |
| Transcripts, part 2 | Everything with captions or audio on BlackRock's Aladdin and eFront media library, the Never Done and Aladdin Client Stories podcast feeds, the Build vs Buy webinar and client videos | 95 files |
| Job postings and press releases | Studio Developer, Technical Writer and other Aladdin postings; BlackRock releases on Azure, AWS, Preqin and eFront, as captured in the research briefings | as cited |

## How to read the source tags

Every fact ends with a tag in square brackets. `[spec: OrderAPI]` means the public API specification of that name. `[SDK README]` means the AladdinSDK README on PyPI. `[page: Copilot]` means a BlackRock web page. `[T1-11 ...]` and `[T2-049 ...]` mean transcript files in part 1 and part 2 of the transcript set, with the file number and a short title. `[briefing: suite architecture]` means one of the two research briefings built from the specifications and BlackRock's pages. Quotations are verbatim from the transcript files; where a file is an automatic caption, the tag says so. Caption misspellings of proper names inside quotations (for example BlackRock and Preqin) were corrected; no other wording inside quotation marks was changed.

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# Part 1. Aladdin

## 1.1 What Aladdin is

Aladdin is the investment platform BlackRock built for its own business and sells to other institutions. BlackRock's own description, as printed at the foot of its eFront pages, is: "Aladdin® is a technology platform used by asset managers and owners including banks, financial institutions, pensions, corporations, insurers, and wealth managers. The Aladdin platform unifies the investment management process, providing a common data language within an organization to enable scale, provide insights, and support business transformation." [page: eFront Insight LP video] The same text adds that "Aladdin® and eFront® platforms bring together public and private asset class data to create an industry-recognized whole portfolio solution for investment professionals globally." [page: eFront Investment Café video]

Brennan Rosales, AI engineering lead, gave the engineering-side version at LangChain Interrupt 2025: "Aladdin is a proprietary technology platform that unifies the investment management process. It provides asset managers with a whole portfolio solution that gives them access to both public and private markets and allows them to cover the needs of both institutional and retail investors. This system is built by the industry for the industry. BlackRock is the biggest user of Aladdin." [T1-16 LangChain Interrupt, 2025]

The "whole portfolio" framing is BlackRock's current product language. The Whole Portfolio page describes "a connected ecosystem that gives institutional investors an enterprise-wide view across all asset classes—including both direct and indirect investments in private markets—enabling unified risk, exposure, as well as performance analysis and strategic oversight", and lists "an integrated ibor that supports public and private assets" among its three pillars. [page: Whole Portfolio] The page's counters read "6 years delivering Whole Portfolio solutions—with 20+ years delivering solutions for private markets" and "50,000 unique private markets companies maintained and 2k+ private markets risk factors monitored daily". [page: Whole Portfolio] Tarek Chouman, Global Head of the Aladdin Client Business, in the APAC client forum recap: "BlackRock and Aladdin was built for the concept of the whole portfolio." [T2-063 APAC Client Forum recap, 2025]

The investment book of record is the anchor. Kunal Khara, Global Head of Aladdin Product: "Our iBOR is comprehensive and consistent that allows all clients and all users to have a single source of data for their portfolio, for their positions, for their exporters." [T2-061 Best Overall Technology Provider 2023, interview 2024] Richard Newman, GM of Aladdin Studio: "It does start with IBOR. This other IP we have here at Aladdin that we create, data around credit analytics, around mortgage-backed security data, around insurance analytics, things linking that into the Aladdin Data Cloud." [T1-11 Never Done: Data, the key ingredient, Jan 2025] The specs model both books: AborLot records carry a parallel OBOR lot id so the accounting book and the investment book reconcile at lot level. [spec: AborLotAPI] [briefing: suite architecture]

Who uses it inside a client firm, as the transcripts describe them: portfolio managers, traders, trading and data operators, compliance, risk, performance analysts, fund accountants and CFOs. Dave Woodhead, head of engineering for Aladdin Studio: "the Aladdin application is a suite of tools that power the entire investment life cycle from a portfolio manager that's managing a portfolio ... The portfolio manager is the tip of the iceberg. Behind them, you have an entire raft of data operators of of trading operators that make sure that the investment life cycle works as it's supposed to." [T1-01 Snowflake Data Cloud podcast, Woodhead, about 2024] The Aladdin Accounting team: "Our clients are CFOs, they're controllers, they're heads of performance". [T1-33 Big Little Questions: Accounting] The UserAPI classifies accounts as a client employee, "A User who is an employee of a AladdinProvider", a system user, or a BlackRock employee. [spec: UserAPI]

## 1.2 Where it came from

Woodhead dates the platform at "just over 20 years" and says BlackRock "was founded on the notion that technology is a differentiator ... As as BlackRock grew, our clients recognized that our technology was differentiated and asked could they use the same software. So BlackRock pretty quickly entered the fintech business and we started offering Aladdin as a service." [T1-01 Snowflake Data Cloud podcast, Woodhead, about 2024] Randy Fradin, in a Big Little Questions episode on the Azure migration, says "A Latin was created, you know, 30 years ago. And in that time, we've demonstrated repeatedly our ability to kind of completely re-underwrite the stack of technology and capabilities underpinning a Latin. I think our ability to do that without disrupting customers is kind of our superpower." [T1-28 Big Little Questions: platform and solution, Fradin and Kerner, undated] The Fivetran write-up says Aladdin "has served as the backbone of BlackRock's technology and investment operations since the company's early days". [page: Fivetran blog, Aug 2024]

The engineering history is told in one BlackRock podcast, the Principles of Design episode with Sue Zheng, Tech Fellow, and Afwa Kandawire, Global Head of Aladdin Platforms (May 2023). Zheng on the founding rule: "one of the engineering ethos we always had was one database. You join, people tell you it's one database, you cannot create another database. But at some point, the company continued to grow and the industry grew, you realize there's a physical constraint to one database. So, then we evolved one database to one source of truth". [T2-027 Principles of Design, May 2023] On the first break from it: "when we first started to challenge that in trading, we were trying to figure out how to keep up with the growing trading volume and the need for more speed without impacting the rest of the system. And we realized that we probably have to take parts of the order management or order execution system and put it on top of a different real time transactional database ... at that point we leveraged Cassandra. So, before that everything was in one database, a side base database." [T2-027 Principles of Design, May 2023] "Side base" is the transcript's rendering of Sybase. The technique was "writing behind, so that yes, we're going to be in a separate database so that we can reduce the contagion risk, but also making sure everyone still has access to the same information no different than before." [T2-027 Principles of Design, May 2023]

Zheng on Charlie Hallac, whom Kandawire names as one of the founders of that thinking: "He would check every database change. You want to make sure that again, we stick to the one database. If you're going to write to this, make sure there's only one application writing to it ... The service was just called a BMS service back then, but it's the same concept." [T2-027 Principles of Design, May 2023] She also describes rebuilding "an old C++ application into Java" and stripping compliance checks out into a shared service. [T2-027 Principles of Design, May 2023]

Kandawire on the current transition: "if we've been in this architecture for almost two decades over the last few years, we've started to migrate away from our layered architecture to a services architecture ... Every single client environment looks the same, is built the same way, we have a single control plane and a single source of truth. And so, you know, one of the things that we're starting to introduce is this notion of domain driven design and the idea being that the database is no longer the single source of truth, but rather the domain is the single source of truth for that domain." [T2-027 Principles of Design, May 2023]

On the same-code rule, Zheng: "in Aladdin we always try to push the same software out to all clients ... I don't want to be on the hook for saying I have like 50 different versions out there." Kandawire: "That notion of a single code base at all clients leads to simplicity, it's what enables the scale, it's what enables the reliability." Zheng: "BlackRock, the house, is using the same software, right? We have skin in the game". [T2-027 Principles of Design, May 2023]

On the cloud decision, Kandawire: "I arrived, I joined at a point in time where we had made a decision on how we were going to migrate to the cloud. We had chosen our cloud provider. And one of the core, as you know, principles around the migration is that the first stage would be a lift and shift, and that meant really focusing on moving the infrastructure ... the principal reason for doing a lift and shift was to ensure that we could actually

enable Aladdin to be available to more prospects in more regions than we had datacentres available." [T2-027 Principles of Design, May 2023] Microsoft's Ignite 2024 recap (Dec 2024) dates the start: "Five years ago, we began working on a strategy to move them into the cloud on Microsoft Azure. Today, all BlackRock's technology platform clients and services are hosted in the Azure cloud". [page: Microsoft Cloud blog, Dec 2024] Lance Braunstein, head of Aladdin Engineering, in March 2023: "you're great at data centers. We will never be the data center operator that you are, and that's fine. So, we have partnered with you." [T2-022 Microsoft partner story, Mar 2023] The Studio briefing adds, from the Microsoft customer story and the AWS release, that the migration was client by client with Microsoft Treasury the first instance moved, and that in December 2025 BlackRock announced an AWS hosting option, "cloud-agnostic by design", with general availability for AWS-hosted clients expected in H2 2026. [briefing: Studio]

The second origin thread is private markets. Sudhir Nair, Global Head of Aladdin: "Aladdin grew up very much as a public markets oriented technology. And it wasn't until 2019, when we did our first major technology acquisition with E-Front, that we had the opportunity to bring side by side through this concept of the whole portfolio, the public markets investment technology alongside private markets investment technology." [T2-033 Never Done: Sudhir Nair, Jun 2025] In the same episode he says "about three months ago, we closed our next acquisition", Preqin. [T2-033 Never Done: Sudhir Nair, Jun 2025]

## 1.3 What it does: the investment lifecycle it covers

Lou Rosato, speaking at Symphony Innovate 2026 for BlackRock operations: "We have a technology called Aladdin. That's our proprietary workflow. It goes end-to-end from risk to investment to trading to post-trade to accounting all the way through the life cycle." [T1-10 Symphony Innovate 2026] Khara: "a lot of firms very much value this integrated end-to-end investment management process, the front middle and back office all managed through Aladdin." [T2-061 Best Overall Technology Provider 2023] The Aladdin Accounting team: "what we've built is a process where you can go from that first order through the trade, through all the operation protocols into accounting without lifting up the pencil". [T1-33 Big Little Questions: Accounting]

The 54 public specifications are organised into eleven domains: accounting, analytics, clients, compliance, data, investment-operations, investment-research, platform, portfolio, portfolio-management and trading. [briefing: suite architecture] The specs show pre-trade compliance as a property of the order itself: posting an order takes `whatIf` ("True to post orders in a what-if mode") and `runCompliance` ("Default is true to run compliance on this post order request") flags and returns violations inline. [spec: OrderAPI v1] Compliance rules come in ten types (prohibition, concentration, value at risk, disclosure, scripted, look-through definition, information, trade, counterparty, counterparty exposure limit) with overnight and intraday run flags. [spec: ComplianceRuleAPI] Cash forecasting is a settlement-date ladder per currency that can include intraday trades, authorised and open orders, confirmed and estimated new cash, collateral pledged and held, and unfunded commitments. [spec: CashLadderAPI v2] Operations APIs cover corporate actions and entitlements, coupon resets, MBS factors, miscellaneous cash transmissions, collateral statements and broker and settlement reference data. [briefing: suite architecture] The accounting domain is described in its own spec as "Aladdin Accounting". [spec: PortfolioConfigurationAPI]

Performance and reporting run through Explore. A BlackRock client training recording: "performance analysts primarily leverage explore a One-Stop shop for reporting and portfolio analysis". Returns "are calculated at the security level and aggregated to the sector portfolio and Enterprise levels daily", decomposed into price, income, pay down, FX and trade P&L, with "a Brinson 2 or three Factor attribution analysis" over a risk-based decomposition. [T1-06 Fixed Income Performance Analyst, undated] The Explore classroom recording describes a web application launched "through your genie toolbar" or by adding "/app/Explorer" to the client site URL, with workspaces, widgets, risk settings ("analytical var as a 95% instead of an 84%"), stress P&L columns, what-if models and saved what-if portfolios; the risk summary tab is "pulling straight from PRT". [T1-09 Aladdin Explore

Intro, undated] In a BlackRock webinar Explore is called "One of the main kind of all flagship tools that we have available ... This is our portfolio analysis and reporting application." [T2-048 Assessing the impact of escalating trade tensions, undated]

The table maps lifecycle stages to the applications the specs and transcripts name.

| Lifecycle stage | Aladdin applications and internal systems named | Source |
|---|---|---|
| Research and idea capture | investment-research domain: ResearchNote, Engagement, AnalystCoverage, Watchlist, Criterion (all alpha) | [briefing: suite architecture] |
| Portfolio setup and mandate | PortfolioToolkit (850-attribute Portfolio entity), "portfolio monitor" seen by PMG ("PMG Monitor Horizon Days"), Green Package risk services, FX Center (accounts in or out of scope "for making FX payments via the FX Center") | [spec: PortfolioToolkitAPI] |
| Client custom data | CID Manager UI, which uploads entity files to Custom Investment Datasets | [spec: CIDS writer] |
| Cash | CashMaster ("Cash publishing is done using an Aladdin application called CashMaster"); AlphaCashMaster and AutoMatchServer for committed cash | [spec: CashLadderAPI v2] [spec: PortfolioToolkitAPI] |
| Order raising and compliance | OrderAPI with whatIf, runCompliance and complianceTimeout; "Order Monitor of Aladdin Dashboard"; order statuses "Open", "Compl Pending", "Authorized", "Active"; Abacus compliance engine in parallel or live mode | [spec: OrderAPI v1] [spec: ComplianceRuleAPI] |
| Trading and execution | TradeAPI v2 block and child trades, quotes in competition, external trade references | [spec: TradeAPI v2] |
| Risk and oversight | Explore (portfolio analysis and reporting), PRT risk reports, Risk Radar (risk rules, exceptions, workflow, config) | [T1-09 Aladdin Explore Intro] [T2-048 trade tensions webinar] [spec: RiskConfigAPI] |
| Corporate actions | "Dashboard Events Monitor", on which an approved event appears and later falls off | [spec: CorporateActionAPI] |
| Coupons and factors | Bulk Rate Mechanism (BRM) for internally generated coupon resets | [spec: CouponResetAPI] |
| Collateral and margin | DECO (legal agreements; statements "viewable in the DECO tool"); Paycheck (failed swap payments) | [spec: CollateralStatementAPI] |
| Payments out | the "transmission framework", which sends money out once confirmedBy and reviewedBy are recorded | [spec: MiscellaneousCashflowAPI] |
| Capital flows | "newcash" module and VIEW_NEWCASH permissions | [spec: EnrichedCapitalFlowAPI] |
| Accounting and NAV | Aladdin Accounting: ABOR lots, valuations with sign-off, composite membership | [spec: AborLotAPI] [briefing: suite architecture] |
| Performance and attribution | Explore | [T1-06 Fixed Income Performance Analyst] |
| Data distribution | ADC Admin Center (Studio) for Aladdin Data Cloud datasets | [spec: AdcDatasetAPI] |
| Private markets | eFront Insight, Investment Café, Invest GP, Invest LP, Invest AS, FAIR, eFront Provider, eFront Office | [page: eFront site navigation] [T2-087 Meridiam] |

Positions, risk analytics outputs, performance attribution, pricing, benchmarks, execution management and private markets are absent from the public specification packages. [briefing: suite architecture]

## 1.4 The product family

The Aladdin site's navigation lists four platforms (Aladdin, Aladdin Wealth, eFront, Preqin) and nine solutions (Accounting, Aladdin Studio, Aladdin Data Cloud, Whole Portfolio, Risk Management, Sustainability, Climate Risk, Aladdin Copilot, Private Markets Data). [page: Copilot] The name "Aladdin Enterprise" does not appear in the pulls.

| Product | What the corpus says it is | Source |
|---|---|---|
| Aladdin (core platform) | The end-to-end platform described in 1.1 and 1.3. The Whole Portfolio page calls it "an open, API-first platform that adapts to your code and strategy across the investment management process" with "an ecosystem of partners across data vendors, systems integrators, and liquidity providers". | [page: Whole Portfolio] |
| Aladdin Risk | The analytics core. The Studio for Risk page: "Aladdin® Studio extends Aladdin Risk beyond the user interface, helping organizations embed risk analytics and data into the systems and workflows where their teams already work." Liz, head of transformations at Aladdin Wealth: "the Aladdin risk model, which was part of BlackRock's founding years ago ... it's a 3,000 risk factor model". | [page: Studio for Risk] [T1-08 iCapital Connect, Aladdin Wealth] |
| Aladdin Wealth | "BlackRock's proprietary technology platform that enables wealth managers to accelerate growth across client segments ... combines sophisticated risk analytics with comprehensive portfolio management on a single, unified platform." Ted Stratigos is "Global Head of Aladdin Wealth Tech". Morgan Stanley's Portfolio Risk Platform "leverages a combination of BlackRock's Aladdin risk analytics and Morgan Stanley data". Intesa Sanpaolo's Fideuram Direct "is based on the Aladdin platform and has a model of advisory based on Aladdin". | [page: Wealth Auto Commentary, 2 Oct 2025] [T2-002 Intesa Sanpaolo, Mar 2024] |
| Aladdin Climate | Climate analytics built on scenarios "provided by the network for green and the financial system and GFS and the intergovernmental panel for climate change, IPCC", translated into cash-flow impacts and "the Aladdin traditional asset evaluation models"; coverage "around 3 million securities for about 9,000 ultimate holding issuers" across equity, fixed income including sovereign and securitised, funds and private market securities. The ClimateAPI exposes physical, transition and temperature-alignment metrics under named scenarios. | [T2-047 An introduction to Aladdin Climate, undated] [spec: ClimateAPI] |
| eFront | Acquired 2019, "our first major technology acquisition". The eFront site covers "alternative investment data management ... From fundraising, fund administration and reporting to data services and analytics". Products: Insight, Invest GP, Invest LP, Invest AS, Investment Café, FAIR, eFront Provider. Investment Café: "more than 10,000 funds choose to report with e-front investment cafe, securely sharing over 10 million documents through the portal." Tarek Chouman "was the CEO of eFront"; founder Olivier Dellenbach. | [T2-033 Nair, Jun 2025] [page: eFront site] [T1-35 eFront Investment Café video] [T2-014 Democratizing private markets, Sep 2025] |
| Preqin | Private markets data company founded by Mark O'Hare, "the home of alternatives", with "hundreds of researchers" and data collection "from 4 million news articles a day". Closed about March 2025 per Nair ("about three months ago" in June 2025); BlackRock welcomed "the 1600 colleagues from prequin". Nair in Jan 2026: "We did that with Preqin this year." The Sept 2025 webinar: "investors now have access to over 15,000 fund investments within e front insight" and Preqin "is being embedded into e front insight"; the Studio briefing records the eFront Insight integration in Feb 2026. | [T1-07 Preqin video] [T2-033 Nair] [T2-062 Preqin completion video, 2025] [T2-016 Rethink Operating Models, Jan 2026] [T2-056 Unifying the Whole Portfolio, Sep 2025] [briefing: Studio] |
| Aladdin Provider | A network connecting asset managers to asset servicers. Chouman: "we developed the Aladdin provider network to bring a greater connectivity between asset managers and asset servicers ... Now we've expanded our provider network to include nine of the largest global asset services". J.P. Morgan "was one of the very early adopters ... from 2016". | [T2-059 JPM client story, undated] |
| Aladdin Explore | The portfolio analysis and reporting application (section 1.3). | [T1-09] [T2-048] |
| Aladdin Copilot | Generative AI assistant "now available to all Aladdin clients" (section 1.8). | [page: Copilot] |

| Product | What the corpus says it is | Source |
|---|---|---|
| Aladdin Data Cloud | "part of Studio" per Newman; Snowflake-based (section 1.7 and Part 2). | [T1-11 Newman, Jan 2025] |
| Aladdin Studio | The extensibility layer: APIs and events, ADC and developer tools (Part 2). | [T1-01 Woodhead] |

## 1.5 How it is built and run

Scale of the organisation, per Rosales in 2025: "we also sell and support hundreds of clients globally across 70 countries all using the Aladdin platform. The Aladdin organization is made up of around 7,000 people, with over 4,000 of those being engineers. And these engineers build and maintain the around 100 Aladdin front-end applications that thousands of people use every day." [T1-16 LangChain Interrupt, 2025] There are "say, 50, 60 Aladdin engineering teams that cover specific domains throughout the platform". [T1-16 LangChain Interrupt, 2025]

Tenancy and control plane. Kandawire: "Every single client environment looks the same, is built the same way, we have a single control plane and a single source of truth." [T2-027 Principles of Design, May 2023] The specs carry the tenant everywhere: `clientAbbreviation` on ADC datasets, `accountCode` on external trade references, a per-client ADC database, reconciliation of one client's datasets against a gold-copy client, OAuth endpoints at client.blackrock.com, and user types that separate client employees, AladdinProvider employees, system users and BlackRock employees. [spec: AdcDatasetAPI] [spec: TradeAPI v2] [spec: UserAPI] [briefing: suite architecture] Kandawire's platform organisation runs "product families": mission control (keeping Aladdin up and maintenance "with a cadence and in a highly controlled way"), hosting services, engineering services, enterprise architecture and a product office. [T2-027 Principles of Design, May 2023]

Hosting. Azure since the lift-and-shift described in 1.2; "all BlackRock's technology platform clients and services are hosted in the Azure cloud" as of December 2024, with "New rollouts now take weeks instead of quarters to complete" and data "encrypted at rest and in transit". [page: Microsoft Cloud blog, Dec 2024] Fradin: "Initial step of moving the Latin into Azure was always meant to be the springboard to what comes next, which is essentially a perpetual cloud journey." [T1-28 Big Little Questions: Fradin and Kerner] Matthew Kerner of Microsoft in the same episode notes Aladdin "is now hosted in this cloud, where there's this deep partnership with OpenAI, where you have the Azure Machine Learning capabilities and the ... Azure OpenAI integration". [T1-28] AWS: announced December 2025, availability for AWS-hosted clients expected H2 2026. [briefing: Studio] Khara: "a lot is built on a very, very scalable cloud-based infrastructure and architecture to handle the heightened volume, as well as the variety and the velocity of data and transactions". [T2-061] Job postings describing an Aladdin Container Platform on Kubernetes with a Go control plane are reported in the Studio briefing at lower confidence; the specs themselves are hosting-neutral. [briefing: Studio] [briefing: suite architecture]

What the specifications show about engineering conventions:

- Protobuf and gRPC underneath, REST on top. Every spec carries `rpcStatus` (google.rpc.Status), `Any`, `google.type.DateTime`, `updateMask`, `pageToken`, `LongrunningOperation`, enums named `FOO_UNSPECIFIED`, and message names such as `v1FilterOrdersRequest`: the fingerprint of services defined in protobuf and exposed through gRPC-gateway style transcoding. "Aladdin Graph" names the unified layer; specs cite naming "agreed with Aladdin Graph team". [briefing: suite architecture] [spec: OrderAPI v1]
- Google API design conventions enforced by a linter, with inline suppressions such as `api-linter: aladdin::9016::query-custom-method-http=disabled aip.dev/not-precedent` and the justification

"changing existing URI to suggested URI will break all citizen developers code". [spec: RiskRuleAPI] [briefing: suite architecture]

- Stability labels: 24 alpha, 6 beta, 23 stable, 1 sunset across the 54 public specs; 190 of 301 operations are POST; 35 operations are flagged long-running. [briefing: suite architecture]
- Bitemporal fields: entities such as RiskException and RiskCustomEvaluationMetric carry `validBeginDate`, `validEndDate`, `entryTime` and `expiryTime`; ADC datasets have a `bitemporal` property. [spec: RiskExceptionAPI] [spec: AdcDatasetAPI]
- Four-eyes approval: ADC dataset states are DRAFT, REVIEW ("dataset is ready for a 4-eyes check prior to release"), APPROVED and REJECTED; compliance rule assignments carry `resolutionRequiresFourEyesApproval`; cash transmissions require `confirmedBy` and `reviewedBy`. [spec: AdcDatasetAPI] [spec: ComplianceRuleAssignmentAPI] [spec: MiscellaneousCashflowAPI]
- Orders and compliance: `complianceTimeout` is "specified in seconds (example format: "10s") ... If the non-zero time is reached and the run is not complete, orders will be posted with the "Compl Pending" status set regardless of any violations". [spec: OrderAPI v1] Trades carry a `compliancePending` flag. [spec: TradeAPI v2]
- Two-layer authorisation. Every call carries an application key header `VND.com.blackrock.API-Key` plus OAuth 2.0, a client key or basic auth, with scopes per resource such as `trading.order_management.order.v1.Order:read`; inside the service, legacy permission codes still apply: `brkrMaint` for broker maintenance, `asReAsMOD` for restricted assets, `PortGroupMOD`, `asPortMOD` and `GPMaint` for portfolio groups, `PERMS_GROUP_MOD_perm` or `MOD_PERMS` for permissions, and the newcash set `VIEW_NEWCASH, CASH_PAY, CASH_MOD, CASH_CONF, ASSET_MOD, ASSET_CONF, CASH_TRANSFER, ASSET_TRANSFER`. [spec: BrokerAPI] [spec: RestrictedAssetAPI] [spec: PortfolioGroupAPI] [spec: UserGroupPermissionAPI] [spec: EnrichedCapitalFlowAPI] [briefing: suite architecture] The platform user and permission APIs state "This is not intended to be used for Authorization." [spec: PermissionAPI]
- Internal expression languages: `bqlExpression`, `bqlFilter` and `jql` fields on trade routing. [briefing: suite architecture]

## 1.6 Risk and analytics

Risk is the historical core. Woodhead: "at our roots where we started, our differentiation was our ability to understand the risk of the market. That's what we call our analytics factory ... We called it the green package. So every day a portfolio manager called it the green package because it was printed on green paper ... That's my portfolio risk for the day." [T1-01 Woodhead] The Green Package survives in the specs: a portfolio attribute "Indicates whether or not risk services/green package are part of Client Services" and a compliance disaggregation option "Green Package Setting". [spec: PortfolioToolkitAPI] [spec: ComplianceRuleAPI] Woodhead adds that start-of-day risk "is still the heart of our of our data ecosystem. It's the most valuable product that comes out of our analytics factory." [T1-01]

Factor counts stated in the corpus: "2k+ private markets risk factors monitored daily" [page: Whole Portfolio]; "a 3,000 risk factor model" for Aladdin Wealth [T1-08 iCapital Connect]; "my 2500 plus factors" in the Explore stress-test demo [T2-048]; "We have factor models across our alternatives and our public assets" [T2-048]. The fact-check attributes "monitors over 5,000 risk factors daily" to the Fivetran write-up, but the Fivetran blog text in the pulls does not contain that phrase; the 5,000 figure is not found in the pulls. [page: Fivetran blog]

Scenario analysis: BlackRock gives clients "a stress scenario as a basis or as a viewpoint" and users can "add multiple scenarios where you can also just edit these scenarios shocks", select a different risk model and apply long- or short-term settings; the demo scenario is "the US euro decoupling scenario". [T2-048] The Whole

Portfolio FAQ lists "proprietary risk models, stress testing tools, pacing analysis, and real-time analytics". [page: Whole Portfolio] Yulia Yudelevich and Michael Sternberg, on the risk models: "what we try to do with our risk models is to translate data into actionable information that allows our clients to make investment decisions ... There are a lot of new forms of data, unstructured data, news events ... And we need to bring that into our analytics and make that available to people, but with caveats." [T1-22 Big Little Questions: Analytics and risk, 2024]

Risk governance in the specs is a separate framework from compliance: RiskRule (metric, limit and breakdown, "250 bps tracking error" style absolute or relative limits), EvaluatorAnalytics (metric source "Aladdin Analytics" or custom data), RiskCustomEvaluationMetric, RiskException and RiskWorkflow, and RiskConfig that drives both evaluation and "the Risk Radar UI". [briefing: suite architecture] [spec: RiskConfigAPI] Analytics is exposed to clients as data retrieval and rule evaluation, not as a calculation engine they call directly. [briefing: suite architecture] Newman, January 2025: "we have these engines that historically were run through interfaces. So we're providing those APIs now. We're releasing soon the ability to actually access those analytic APIs, portfolio analysis, security analysis, things like that. That's all part of the Studio umbrella." [T1-11 Newman, Jan 2025] The Studio for Risk page: "Run analytics programmatically: Compute portfolio and security analytics for repeatable reporting, analysis, modeling, and integration into downstream systems." [page: Studio for Risk]

Performance and attribution: security-level daily returns, P&L decomposition, risk-based decomposition fused with Brinson attribution, ex-post statistics (tracking error, beta, information and Sharpe ratios, upside and downside capture, batting averages), batch reports to Excel or PDF, all in Explore. [T1-06 Fixed Income Performance Analyst] Climate analytics: see 1.4; the ESG API normalises multi-vendor measures (examples "MSCI, SUST, ISS, REFINITIV, CLARITY_AI") into one schema. [spec: EsgDataAPI] Jess Fitzgibbon, Head of Product for Aladdin Risk, in July 2026: "We've moved beyond the spreadsheet into the more exception based workflows and now we're moving into beyond exception based workflows." [T2-028 Reducing time to insight, Jul 2026]

## 1.7 Data

Woodhead's phrase for the data model is "the language of the portfolio": "when you hear us speak about the language of the portfolio, what we mean is we're stitching together all of the constituent pieces that make the investment process work." On symbology: "Getting that data is very easy. Mapping that data and understanding that in the context of the investments that you've made is very very hard. It requires you to do a lot of symbology processing. How do I bring the data into a common language? ... we bring all of that together into a single symbology, a single language, the language of the portfolio". [T1-01 Woodhead] The ACF 2026 opener: "So we built a common language for portfolios." [T2-060 Aladdin Client Forum 2026] Kamya Somasundaram on trading connectivity: "that is how Aladdin, for example, communicates to a network of hundreds of participants, broker dealers, venues, trading platforms." [T2-017 Aviva Investors, Dec 2025] The specs address securities by Aladdin asset id, CUSIP, ISIN, SEDOL, RIC or CINS, and the SecurityCreation API writes CDS, CDX, equities, options, futures, FX instruments, swaps, swaptions, time deposits, repo and MBS TBAs. [briefing: suite architecture]

Before Aladdin Data Cloud, Jeff Miller, Managing Director, Global Head of Data Platform and Product Engineering and Chief Architect: "Prior to our adoption of Snowflake and the advent of our Aladdin Data Cloud offering, like many firms, we used legacy and traditional ways of distributing data to our customers. We sent flat files, we sent large schema definitions, FTP and related transfers". [T1-17 Snowflake: Jeff Miller, undated] Woodhead: "it was around 2019 we started look understanding that our clients are building their own data ecosystem ... we didn't have a data platform that could do that at the time and that's where we started our our conversation with Snowflake." [T1-01] The Fivetran write-up: ADC "is supported by Fivetran for data movement and Snowflake as its cloud data platform"; users access "their Investment Books of Record (IBOR), Accounting Books of Record (ABOR) holdings and other market data". [page: Fivetran blog, Aug 2024] Scale: "we operate um hundreds of Snowflake accounts. We share data to hundreds of Snowflake accounts and we're managing hundreds of

vendors, tens of thousands of pipelines." [T1-01] Diwakar Goel, Global Head of Data: ADC is "a data integration hub so that you can access all the data that you need in a lot in an e-front through one single source within 15 minutes of latency from various kinds of sources". [T2-050 Data as a strategic asset for pensions, undated] Daily data volumes in bytes or rows are not stated in the pulls. Part 2 covers Data Cloud, Studio and the SDK.

## 1.8 Artificial intelligence in Aladdin

Aladdin Copilot, per its product page: "Introducing Aladdin® Copilot, now available to all Aladdin clients. Through generative AI, Aladdin Copilot serves to strengthen the connective tissue across the Aladdin® platform, surfacing answers instantly to support key business decisions". It "Will not give investment advice or respond to questions outside Aladdin platform boundaries" and "Contains content filtering and parameters to limit risk of hallucination, misinformation or inappropriate outputs", with "permission-dependent access to insights". [page: Copilot] eFront Copilot is "now available within eFront Insight" and "built on the Aladdin-isolated instance of Azure OpenAI-secured connection where all personal information is filtered out". [page: eFront Copilot] Microsoft's Dec 2024 recap: "with the launches of the Aladdin and eFront copilots, they are now delivering a set of natural language interfaces ... BlackRock technology clients are already using this capability today", and BlackRock started "an AI Lab in 2018". [page: Microsoft Cloud blog, Dec 2024] Khara, in the awards interview recorded for 2024: "on the E-front side in particular, we have a co-pilot that is out in the market very soon, there'll be something we're going to be doing on the Aladdin front as well ... a big, big, big focus area for us in 2024." [T2-061]

The architecture, from Rosales and Pedro Vicente Valdez, Principal AI Engineer, at LangChain Interrupt 2025: Copilot "is currently embedded across all of those 100 front-end applications" and "acts as the connective tissue across the Aladdin platform". The plug-in registry lets the "50, 60 Aladdin engineering teams" onboard in two ways: "we come to define a tool as kind of mapping one-to-one to an existing Aladdin API already out in production" or "spin up a custom agent and also plug that into the system". Design started "two, two and a half years ago" with "a very slim, very specific version" of an agent protocol, and the team is "actively evaluating" LangChain's agent protocol and A2A. A query enters "our orchestration graph. From day one, it's been built using Langchane": an input guardrail node for "off topic and toxic content" and PII, then a filtering and access control node over "couple of thousands agents and tools that are registered in the plug-in registry", where teams control "which environments this plug-in is enabled in, which user groups have access to this, and even which Aladdin apps can even access these plug-ins"; the set is cut to "say, 20 to 30 tools and agents" because "if we're sending more than 40, 50 tools, we probably won't have that good of a performance". Orchestration is "very reliant on GPT-4 function calling today", iterating planning and action, then output guardrails to "detect hallucinations as best as we can". [T1-16 LangChain Interrupt, 2025]

Valdez: "We're all using supervisor ... And we are just one more of those presentations using supervisor. Why? Because it's very easy to build. It's very easy to release. It's very easy to test." On evaluation: "you have to do evaluation-driven development ... if I write, you're a lot in copilot, you must never provide investment advice. What is investment advice? I need to generate a lot of synthetic data ... and build an evaluation pipeline and make sure that we are never going to give investment advice." The report "runs every day. It runs on every PR", and "for every plugin that comes into a system, we ask them for ground truth". [T1-16 LangChain Interrupt, 2025] Context the system uses: "which Aladdin application they're asking this question from, what they actually have up on the screen, what portfolios, what assets, what widgets". [T1-16]

Human in the loop. Adrian Crockett of Microsoft, on BlackRock's podcast: "making sure that we've got human in the loop on the vast majority of operations, particularly in regulated industries, is going to be really, really critical." [T1-12 Humans in the loop part 1] And: "in order to do that, you need to actually create the plug-ins. There's no copilot without plug-ins." [T1-13 Humans in the loop part 2, Nov 2024] Syril Smith Garson, Head of Product for AI: "we're very focused on how do we build this platform ... and a set of common services and capabilities that underpin all of these different copilot experiences" and "how do we think about Aladdin, right, as also an

ecosystem, where we can now have plug-ins and capabilities across all of our own applications ... That it can act as this connective tissue across the entire platform." [T1-13] Braunstein in March 2023: "I love the idea of a copilot, so I love the notion that like this doesn't go and make decisions and drive processes on its own." [T2-022 Microsoft partner story, Mar 2023] Pavan Pemmaraju, Head of AI Product and Engineering, July 2026: "there's much more of a focus on ensuring that the data model itself is structured for agentic use cases and we're even changing the way that we refactor the way that our products are built in order to achieve that." [T2-028 Reducing time to insight, Jul 2026] Chouman at the APAC forums: "a portfolio manager is going to be able to talk to Aladdin, where agents are going to be able to work together to make decisions, and Aladdin can provide you with suggestions." [T2-063]

Aladdin Wealth Auto Commentary, announced 2 October 2025: "The new 'Auto Commentary' feature turns complex portfolio analytics and client-level investment preferences into concise insights", combining "the Aladdin Wealth platform's robust data and risk analytics, an individual firm's CIO market outlook, and detailed information about the client's portfolio holdings"; Morgan Stanley's Portfolio Risk Platform "will be the first to implement". [page: Wealth Auto Commentary, 2 Oct 2025] A webinar panellist also described an experiment "using generative AI to read portfolio compliance guidelines and match it to either an existing compliance rule Aladdin or if the rule isn't there to create the relevant code for that new compliance rule". [T2-046 Accelerating Innovation webinar, undated]

The NVIDIA GTC session page in the pulls carries no content (JavaScript-only page), so its talk points are not found in the pulls. [page: NVIDIA GTC] [T1-00 INDEX]

## 1.9 Who uses it and at what scale

Segments on BlackRock's site: asset managers, asset servicers, banks and brokers, corporates, insurers, pension funds, private markets, wealth managers. [page: Copilot] Every Aladdin page carries the disclaimer that the platform "is designed for institutional, wholesale, qualified, and professional investor/client use only and is not intended for end investor use". [page: Studio for Risk] Counts stated by BlackRock on air: "hundreds of clients globally across 70 countries" [T1-16, 2025]; "our ecosystem there's hundreds of participants" [T1-25 Big Little Questions: COO and CTO]. Georgette Anderson on insurers: "From an asset management perspective, we have roughly about 500 billion of insurance asset management on the system and 6 trillion of insurance assets that are actually processed through Aladdin." [T2-040 Big Little Questions: insurers, undated] Khara: "across Aladdin, the work we do today in accounting is almost entirely predicated on bringing this data unification for the investment and accounting books of record for an insurance company together." [T2-010 Global Insurance Report, Nov 2023] BlackRock "is the biggest user of Aladdin". [T1-16] Inside BlackRock, "we have 3,000 citizen developers in business engineers that are interacting with all that in using Python or using R". [T1-01 Woodhead]

Named clients in BlackRock-published material:

| Client | What was said | Source |
|---|---|---|
| Royal London Asset Management | CEO Hans Georgeson: "a need on our part to have a single auto management system"; "we are genuinely in a minority of organisations that achieved it largely on time, if not ahead of plan, and we came in under budget ... So yourselves in Aladdin, ourselves in Royal London, and also, of course, HSBC as our partners"; "the client experience in the past was multiple order management systems"; "the implementation of Snowflake that we did as part of the overall programme has been really successful for us". | [T2-058 Royal London client story, Nov 2024] |
| BNP Paribas Asset Management | CTO Fatia-Fatma Balit on the 2016 to 2020 "Back in the Race" programme: "with BlackRock Aladdin in the front to back chain, with Salesforce in the CRM, and transversally DataHub as a data platform. Along with that, we were able to decommission more than 60 applications"; a new product structure delivered "with no new IT tools". | [T2-008 BNP Paribas AM, Feb 2023] |

| Client | What was said | Source |
|---|---|---|
| Intesa Sanpaolo | Paola Papanicolaou: "we decided together with you to do a full adoption of your platform, across the enterprise on asset management, wealth management, et cetera. Across legal entities as well"; Fideuram Direct "is based on the Aladdin platform". | [T2-002 Intesa Sanpaolo, Mar 2024] |
| SLC Management | Candace Shaw: "We're focused on leveraging our foundational investment systems like eFront, like Aladdin, but also capitalizing on data as a key strategic asset". | [T2-025 SLC Management, Mar 2026] |
| Aviva Investors | Kate Hill, Global Head of Trading, on trading market structure and data architecture, with Somasundaram. | [T2-017 Aviva Investors, Dec 2025] |
| TCW | Melissa Stolfi, Global COO: "asset management is really ripe for AI automation. Because on the operational side, it's very process and workflow heavy." | [T2-018 TCW part 2, Jul 2026] |
| University Pension Plan (UPP) | CIO Aaron Bennett on the Canadian "total portfolio approach, one balance sheet type of model". | [T2-034 UPP, Sep 2026] |
| Ingka Investments | Samuel Rundle: "critical for us was to create more transparency ... We can go back and look at the performance attribution. We get input on risks. We can look across portfolios to actually identify risks that we didn't know were there before." | [T2-030 Ingka Investments, Dec 2024] |
| Helvetia | Dominik Gurtner, CTO of Asset Management, featured on the Whole Portfolio page; the video transcript in the pulls is unusable (non-English audio). | [page: Whole Portfolio] [T2-057] |
| Northstar Group | Priya Gopalan, Head of ESG, on ESG monitoring for a Southeast Asia PE and VC firm with "more than $2.6 billion in committed equity capital". | [T2-003 Northstar, May 2024] |
| Columbia Threadneedle, AustralianSuper | CEO Ted Truscott and COO Peter Curtis in conversation with Nair on operating models and digital transformation. | [T2-005, Jul 2024] [T2-006, May 2023] |
| Morgan Stanley Wealth Management | Portfolio Risk Platform on Aladdin risk analytics; first to implement Auto Commentary. | [page: Wealth Auto Commentary] [T2-036, Oct 2025] |
| Microsoft Treasury | First Aladdin environment moved to Azure. | [briefing: Studio] |
| J.P. Morgan Securities Services | Aladdin Provider early adopter from 2016: "gold copy messaging", "real-time bidirectional workflow for asset and cash reconciliation breaks", rolled out "through simple configuration. There has been no customization, no coding". | [T2-059 JPM client story] |
| eFront users | APG, Allianz Capital Partners, Velliv, Mizuho, Bouwinvest, Meridiam, BBH, Alter Domus, BNP Paribas Securities Services and others on the eFront testimonial pages. Meridiam: "We've been using eFront for three years now ... eFront Office as well as eFront Investment Café." | [T2-064 to T2-095 eFront videos] [T2-087 Meridiam] |

Middle-office outsourcing in the corpus: J.P. Morgan lists "middle office outsourcing" among its services and uses Aladdin Provider so "clients can stay in Aladdin, we can stay in our application, but it still has that real-time workflow between the systems". [T2-059] Mark Erickson: insurance assets outsourced "roughly doubled from about 7% of assets to about 15% of assets over the last six to seven years", against "about $35 trillion" managed globally. [T2-010, Nov 2023] A client panellist on the FT webinar: "we can outsource technical debt to Blackrock for instance they take care of it for us". [T2-046] Awards: BlackRock "won three categories in this year's BST Awards, Best iBOR or Investment Book and Record platform, Best Portfolio Analysis Tool and Best Overall Technology Provider for 2023". [T2-061]

## 1.10 What the pulls do not show

- Pricing, fee schedules, contract values and revenue by product: none in the corpus. [briefing: Studio]

- A client count. BlackRock says "hundreds of clients globally across 70 countries" and nothing more precise on air. [T1-16]
- Positions, risk analytics outputs, performance attribution, pricing and market data, execution management, settlement messaging, eFront and wealth APIs: absent from the public specification packages. [briefing: suite architecture]
- The "5,000 risk factors" figure: not in the Fivetran blog text in the pulls. [page: Fivetran blog]
- The NVIDIA GTC talk content: page retrieved without content. [page: NVIDIA GTC]
- The exact Copilot launch date: the Copilot page is undated; Microsoft's recap (Dec 2024) says launches had happened; Khara's interview (early 2024) says eFront Copilot was "out in the market very soon". [page: Copilot] [page: Microsoft Cloud blog] [T2-061]
- The product name "Aladdin Enterprise", daily data volumes, number of Copilot plug-ins beyond "couple of thousands", and the Studio client list: not stated in the pulls.
- Helvetia's statements: the transcript is machine-generated from non-English audio and unusable. [T2-057]

# Part 2. Aladdin Studio

## 2.1 What Studio is

BlackRock's Aladdin site lists Aladdin Studio under "Solutions", beside Accounting, Aladdin Data Cloud, Whole Portfolio, Risk Management, Sustainability, Climate Risk, Aladdin Copilot and Private Markets Data; the "Platforms" list holds Aladdin, Aladdin Wealth, eFront and Preqin [page: Studio for Aladdin Risk, site navigation]. The Studio product page describes the layer as "API-first", groups its examples under three benefit headings, "Create & automate", "Make Aladdin your own" and "Align internally & externally", and names three users: "a professional engineer building new workflow applications, a code-curious portfolio manager extracting investment insights, or a data analyst automating reporting" [page: Aladdin Studio, as recorded in briefing: Studio and the fact-check A1-12]. The examples under the headings are auto-generating orders and managing targets, projecting settlement cash, intraday what-if analysis, custom scores combining reference and proprietary analytics, and regulatory and board reporting [briefing: Studio].

Dave Woodhead, "managing director and head of engineering for Aladdin Studio" as the host introduces him, gives the definition BlackRock's own engineering uses. "Aladdin Studio is essentially the user experience that allows people to think of Aladdin as a platform. It exposes everything Aladdin knows how to do as a set of APIs events. The Aladdin data cloud built on top of Snowflake takes all of that information that's generated through our analytics factory and inside of Aladdin and pushes that into Snowflake. And increasingly we're building a set of developer tools." On the third pillar: "how do I create environments where they can they have sandboxes, research environments, they can start code, they can build applications. They can automate their workflows." [T1-01 From Analytics to Intelligence, Snowflake Data Cloud podcast, undated; auto-captions]. Richard Newman, General Manager, Aladdin Studio, places Data Cloud inside it: "where the data cloud and Aladdin data cloud specifically comes in, which is part of Studio" [T1-11 Never Done: Data, the key ingredient, Jan 2025].

A second page, "Aladdin Studio for Aladdin Risk", states that "Aladdin® Studio extends Aladdin Risk beyond the user interface", lists "Run analytics programmatically", "Automate portfolio oversight workflows", "Build tailored risk solutions" and "Strengthen data governance and controls", and describes a console whose "centralized experience supports the building, configuration, and governance of Studio workflows, including access management and usage monitoring" [page: Studio for Aladdin Risk].

Inside the corpus, the earliest dated BlackRock statement naming Studio is Kunal Khara's winner interview after the 2023 Buy-Side Technology Awards, recorded ahead of 2024, which calls "the Aladdin studio estate" a focus for 2024 [T2-061 Best Overall Technology Provider 2023; quoted in full in 2.10]. The candidate's draft dates the Studio name to fall 2020 from trade press and to February 2021 from the Data Cloud press release; neither document is in the pulls [REVISED draft Exhibit 26].

## 2.2 History

The Data Cloud conversation with Snowflake began "around 2019", when "our clients were starting to ask the question how do we take you know this really high quality data that's coming out of Aladdin and bring that into our own data ecosystem", and BlackRock "didn't have a data platform that could do that at the time" [T1-01 Woodhead]. BlackRock's press release says Aladdin Data Cloud launched in 2021 [briefing: Studio, citing the press release; V8 S102]. ADC is "a next-generation data-as-a-service solution enabled by Fivetran and Snowflake" [page: Fivetran blog, 28 Aug 2024].

The AladdinSDK core package first appeared on PyPI in May 2024 and stood at version 2.0.0b10 in July 2026; the 11 add-on packages were at 2.0.0b3 [V8 Exhibit 12, with only the 2.0.0b10 version independently verified; briefing: Studio]. BlackRock's engineering blog says the SDK was open-sourced to cut low-level setup for "practitioners and citizen developers" with financial expertise and some programming skill [briefing: Studio, citing engineering.blackrock.com].

The phrase "citizen developer" was used by Kunal Khara, Global Head of Aladdin Product, on 3 November 2023: "while Excel continues to be a part of the technology apparatus for a portfolio manager or a research analyst, ultimately, citizen developers are what are driving these organizations today" [T2-010 Global Insurance Report podcast, 3 Nov 2023]. Lance Braunstein had described the internal programme in March 2023: "we call that program broadly citizen development, but it like the title is not important. What's important is the empowerment, making sure that with power comes responsibility. So, there's a control framework around that" [T2-022 Microsoft and Aladdin Partner Story, 7 Mar 2023]. A March 2024 episode expects "python notebooks up on the desks of the traders" within weeks [T2-001 4IR Part 2, 28 Mar 2024; Whisper].

Woodhead dates the shift from spreadsheets to code: "We several years ago crossed the threshold where more of our users are consuming data programmatically through R through Python ... then they are bringing it down to Excel not in terms of megabytes but in terms of the number of queries the number of users are on the platform" [T1-01]. Kamya Somasundaram, Global Head of Aladdin Partnerships & Trading Product Marketing, describes the same demand: "there is a burgeoning community of individuals saying 'I need that in the form of an API. I need that to speak to this.'" [T2-017 How Data Architecture Shapes Modern Trading, 25 Dec 2025].

The Studio briefing records BlackRock technical-writer postings from 2025 and 2026 describing streaming APIs and webhook subscriptions documented with AsyncAPI alongside OpenAPI [briefing: Studio, job postings]; the posting text is not in the pulls.

## 2.3 The APIs: Aladdin Graph

Every public specification carries an id of the form `agraph.<domain>.<segment>.<group>.<version>.<API>` and a base path `/api/<domain>/<segment>/<group>/v1/` [briefing: suite architecture]. The SDK README calls them "Aladdin Graph API's swagger specifications" [SDK README]. The Order API's linter notes say "postOrder is the agreed naming with Aladdin Graph team" and "Design was agreed with agraph team" [spec: OrderAPI]. No transcript speaker says "Aladdin Graph"; the name exists only in the files [review round 1_vs_podcasts].

The 11 public `asdk_plugin_*` packages, version 2.0.0b3, hold 54 Swagger 2.0 specs and 301 operations: 24 alpha, 6 beta, 23 stable and 1 sunset [apis.json; briefing: suite architecture; V8 verified counts].

| Domain | Public specs | Operations | Example APIs | Stability |
|---|---|---|---|---|
| analytics | 9 | 53 | RiskRule (20 ops), RiskWorkflow, RiskException, RiskConfig, EvaluatorAnalytics, RiskCustomEvaluationMetric, RiskTask, Climate, EsgData | 8 stable, 1 sunset (RiskTask) |
| investment_operations | 15 | 74 | CorporateAction, CorporateActionEntitlement, CouponReset, PrincipalInterestFactor, MiscellaneousCashflow, CollateralStatement, Broker and six broker reference APIs, Contact, AccountInfo, Audit | 9 alpha, 4 beta, 2 stable |
| investment_research | 5 | 41 | ResearchNote v2, Engagement, AnalystCoverage, Watchlist, Criterion | 5 alpha |
| platform | 6 | 29 | User, UserGroup, UserGroupMember, Permission, UserGroupPermission, AdcDataset | 4 stable, 2 alpha |
| portfolio_management | 4 | 28 | CashLadder v2, CIDReader, CIDWriter, Strategy | 3 stable, 1 alpha |

| Domain | Public specs | Operations | Example APIs | Stability |
|---|---|---|---|---|
| portfolio | 3 | 23 | PortfolioGroup, PortfolioToolkit, RestrictedAsset | 2 stable, 1 alpha |
| data | 1 | 15 | SecurityCreation | 1 stable |
| trading | 2 | 14 | Order v1, Trade v2 | 1 stable, 1 alpha |
| compliance | 4 | 9 | ComplianceRule, ComplianceRuleAssignment, Level, Violation | 3 alpha, 1 stable |
| accounting | 4 | 8 | PortfolioConfiguration, Valuations, AborLot, CompositeMembership | 2 alpha, 2 beta |
| clients | 1 | 7 | EnrichedCapitalFlow | 1 stable |
| **Total** | **54** | **301** | | **24 alpha, 6 beta, 23 stable, 1 sunset** |

Source for every row: [apis.json; spec headers in digest].

**Request conventions.** The specs follow Google's API design conventions: resources with custom methods such as `:filter`, `:batchCreate`, `:batchValidate`, `:search`, `:undelete`, `:listRevisions` and `:cancel`; AIP-160 style filter strings; `pageToken` pagination; `updateMask`; `rpcStatus` (google.rpc.Status) and `Any` types; enums named `FOO_UNSPECIFIED` [briefing: suite architecture]. POST is 190 of the 301 operations, and filtering is done by POST with a criteria body rather than GET query strings; batch endpoints are capped at 100 per request for broker reference data, 1,000 for restricted assets and 10,000 rows per page [briefing: suite architecture]. Ten specs expose a `GET /longrunningoperations/{id}` endpoint and 35 operations are flagged `x-aladdin-lro`; the SDK polls these [apis.json; briefing: suite architecture; SDK README]. Many entities carry entry and expiry times or valid-begin and valid-end dates, a bitemporal pattern that supports as-was reporting [briefing: suite architecture]. "Four-eyes" approval appears by name on compliance rule assignments and ADC dataset review; cash transmissions need a confirmer and a reviewer, and risk rules carry sign-off authors [briefing: suite architecture; spec: MiscellaneousCashflowAPI].

The Order API's `v1PostOrderConfig` carries `whatIf`, `runCompliance`, `accountCode`, `externalEntityType` and `complianceTimeout` [spec: OrderAPI, digest]. Posting an order returns compliance violations inline, so pre-trade compliance is a property of the order API [briefing: suite architecture]. The candidate's V8 document, whose add-on figures were recomputed and matched, records `runCompliance` defaulting to true and an order status of "Compl Pending" while compliance runs [V8, OrderAPI definition file]. The Portfolio entity in PortfolioToolkit has 850 properties, with "slices" that expose subsets [spec: PortfolioToolkitAPI, `v1Portfolio | props(850)`]. Fields named `bqlExpression`, `bqlFilter` and `skipBqlValidation`, and a comment "refer to BQL ord_flags", point to an internal query language used for filters and routing rules [briefing: suite architecture; spec: PortfolioToolkitAPI]. Every public spec is tagged `Access.Internal`; the meaning of the tag is not documented publicly [digest; briefing: suite architecture].

The specs carry inline linter suppressions such as `api-linter: aladdin::9016::query-custom-method-http=disabled aip.dev/not-precedent`, which shows BlackRock running Google's api-linter plus its own `aladdin::` rules with justified exceptions [briefing: suite architecture]. In the RiskRule and EvaluatorAnalytics APIs the justification reads: "We need to do this because it is an existing api changing existing URI to suggested URI will break all citizen developers code." [spec: RiskRuleAPI; spec: EvaluatorAnalyticsAPI]. The CashLadder v2 GET endpoint "is provided solely for backward compatibility and should no longer be used" [spec: CashLadderAPI].

**What each domain exposes.**

*analytics.* A risk-oversight suite surfaced in an application the specs call Risk Radar: RiskRule holds metric, limit and breakdown conditions with draft and approved states; EvaluatorAnalytics triggers evaluation of a portfolio

against all or one rule as a long-running operation; RiskException and RiskWorkflow hold breaches and work items; RiskCustomEvaluationMetric lets a client upload its own metric values [apis.json; briefing: suite architecture]. Climate exposes "physical, transition, and temperature alignment analytics as of a given date", and EsgData "a centralized source of ESG data and meta data across multiple vendors" [spec: ClimateAPI; spec: EsgDataAPI].

*data.* SecurityCreation "is used to create CDS, CDX, Equity Equity, Equity Option, Futures, FX NDF, FX SPOT, FX FWRD, FX CSWAP, FX Option, Swap, Swaption, CASH/TD, CASH/REPO, ARM/TBA and MBS/TBA" securities, 15 operations [spec: SecurityCreationAPI].

*trading.* Order v1: "Filter, post or cancel orders. An order is a directive from a portfolio manager to the trading desk to execute a particular investment decision", with `:post`, `:cancel`, `:filter`, `:batchPost`, `:batchCancel` and an order-id filter [spec: OrderAPI]. Trade v2: "Post, cancel, filter and retrieve trade(s)", a BlockTrade with child allocations, broker quotes, settlement details and external trade references [spec: TradeAPI; briefing: suite architecture].

*compliance.* ComplianceRule ("used to automatically monitor whether a fund adheres to a regulation, client mandate, or internal guideline"), ComplianceRuleAssignment (binds rules to portfolios with four-eyes approval), Level (start-of-day and intraday room to limit) and Violation (breaches with contributing positions) [apis.json; briefing: suite architecture].

*portfolio_management.* CashLadder v2 gives "a settlement date based ladder of cash balances for all the exposure currencies for a given portfolio group and cash date"; Strategy names investment strategies that orders and trades are tagged with; CIDReader and CIDWriter load Custom Investment Datasets, where "CIDS does not transform the input data in any kind. The writer of the data owns it" [spec: CashLadderAPI; spec: CIDWriterAPI, via V8 S21].

*investment_research.* ResearchNote v2 (templated notes with permission groups), Engagement (meeting records), AnalystCoverage, Watchlist, and Criterion, stored screening expressions such as `((price > 55 OR oas = 75) AND market_cap < 1mm)` with alert schedules [apis.json; briefing: suite architecture].

*investment_operations.* CorporateAction and CorporateActionEntitlement (elections instructed to the custodian), CouponReset (from "Aladdin's Bulk Rate Mechanism (BRM) module or sourced from external vendors"), PrincipalInterestFactor, MiscellaneousCashflow (which sends money out "through the transmission framework" once confirmedBy and reviewedBy are recorded), CollateralStatement (from agreements set up in DECO), and the broker reference set: Broker, BrokerDesk, BrokerExternalAlias, BrokerConfirmRouting, CounterPartySettlementInstruction and its template, Contact, AccountInfo, Audit [apis.json; V8 S28].

*accounting.* "Aladdin Accounting" configuration attributes by process code, Valuations (NAV workflow with status, sign-off, publication time, Dietz and total return fields, and valuation exceptions with NAV impact), AborLot (lots "retrieved from the ledger_a_summary table" with a parallel OBOR lot id) and CompositeMembership [spec: PortfolioConfigurationAPI; spec: ValuationsAPI; spec: AborLotAPI].

*clients.* EnrichedCapitalFlow: "the cash and asset subscriptions coming into a fund and the cash and asset redemptions going out of a fund", with estimate and actual states, authorise and confirm actions, and permissions from a module called "newcash" such as CASH_PAY [spec: EnrichedCapitalFlowAPI; V8 S30].

*portfolio.* PortfolioToolkit (retrieve, filter and search portfolios and slices), PortfolioGroup (a hierarchy with ancestor and descendant queries), RestrictedAsset ("used prevent assets from being traded in Aladdin") [apis.json].

*platform.* User, UserGroup, UserGroupMember, Permission and UserGroupPermission for administration, which the specs say are "not intended to be used for Authorization", plus AdcDataset, which "Manages Datasets in Aladdin Data Cloud (ADC). Used by Studio's ADC Admin Center." [apis.json; spec: AdcDatasetAPI].

## 2.4 Authentication and authorisation

Every API request carries an application key header, `VND.com.blackrock.API-Key`, for an application registered in Studio, plus one of: OAuth 2.0 authorization code, OAuth 2.0 client credentials, an `X-API-Key` client key, or basic auth [briefing: suite architecture]. The OAuth endpoints are at `client.blackrock.com/api/oauth2/default/v1/` [briefing: suite architecture]. Scopes are per resource, for example `trading.order_management.order.v1.Order:read` and `trading.order_management.order.v1.Order:write` [briefing: suite architecture]. The SDK README says "Client ID and Secret can be sourced from Studio UI by creating a OAuth application under a Project", a basic-auth "API Token" is "registered via Studio UI", and selected scopes "must be a subset of scopes setup for the app in Studio UI" [SDK README].

Authorisation is two-layered. Underneath the OAuth scope, the specs still cite Aladdin permission codes that the calling user must hold, for example `brkrMaint`, `asReAsMOD`, `PortGroupMOD`, `PERMS_GROUP_MOD` and `VIEW_NEWCASH` [briefing: suite architecture]. The capital-flow API requires "standard API permissison ALADDIN_API_USER" and "standard newcash permissions" [V8 S29, S30]. Registering an ADC dataset "requires "StudioAdmin" Aladdin perm assigned to the user", and column refresh names the "StudioAdmin" Aladdin Permission Type and the "DATACLOUD" Aladdin Permission Group [V8 S23, AdcDatasetAPI definition file].

ADC authentication is separate from API authentication. The SDK's ADCClient uses "'oauth' or 'snowflake_jwt' (i.e. RSA Keys) authenticator types for snowflake connectors"; where "ADC OAuth is configured via Studio application, SDK's ADCClient will attempt to fetch an OAuth AccessToken from Aladdin's TokenAPI", on the assumption that the "User has authenticated with the ADC OAuth client application by logging into Aladdin Studio UI" [SDK README].

The User API names three user types: `USER_TYPE_CLIENT_EMPLOYEE` ("A User who is an employee of this client"), `USER_TYPE_PROVIDER` ("A User who is an employee of a AladdinProvider") and `USER_TYPE_SYSTEM`; its description lists "Aladdin Client's employees granted access to Aladdin", "BlackRock employees granted access to client's Aladdin products" and "bot accounts granted access to Aladdin" [spec: UserAPI]. The same description links to a developer guide at `apps/aladdin-developer-portal/#/guides/developer-guide` on the client's Aladdin site [spec: UserAPI].

## 2.5 Events and streaming

Woodhead names events in Studio's first pillar: "a set of APIs events" [T1-01]. Diwakar Goel, introduced on the pensions webinar as Global Head of Data for Aladdin, states the target: "we really want to make every single event that happens in a lot in available to you. So every single transaction that happens should be available and accessible to you via an event, via an API. And that should help you to programmatically build on top of a lot of it" [T2-050 Data as a strategic asset for pensions, undated; Whisper, "a lot in" is Aladdin]. Alisha Sewdass, who "runs our Aladdin Solutions Engineering team for Aladdin in the Americas", lists the same requirement for any investment stack: "So you need access to APIs, you need access to events, and you need to make your workflow much more event based as opposed to batch based" [T2-049 Build vs Buy webinar, undated; published captions].

Inside the public packages, the only event-shaped surface is Studio Notifications. The SDK core package holds "Studio Subscription and Notification APIs" and a `StudioNotification` utility that creates a subscription for recipients and then sends a subject, message and metadata by SMTP or push; a wrapper `send_studio_smtp_notification` gives "a simpler 'email' feature" [SDK README]. The subscription spec lists event types including Studio Space, Studio Compute and ACM [V8 S5]. No AsyncAPI document, webhook specification or event stream is in the pulls; the 54 specs are all OpenAPI request-response definitions [digest].

## 2.6 Aladdin Data Cloud

ADC runs on Snowflake with Fivetran moving the data. "Fivetran allows us to take data from a single source and move that with low latency into a Snowflake landing zone, where we transform it using our enterprise data platform" [page: Fivetran blog, 28 Aug 2024, quoting Woodhead]. Users "can access their Investment Books of Record (IBOR), Accounting Books of Record (ABOR) holdings and other market data", and "users have access to dynamic tables for building Snowpark apps and asking cross-dimensional time-series questions" [page: Fivetran blog]. Fivetran "helps address three key issues: disaster recovery, data recoverability, and consolidation of previously siloed data sets", and the write-up records Woodhead's Snowflake Summit 2024 statement: "We expanded Aladdin® Data Cloud's data network capabilities, enabling clients to bring their own data into ADC, and to distribute Aladdin data to their own data ecosystems" [page: Fivetran blog].

Delivery is by Snowflake data sharing: clients get the data "sitting in their cloud provider of choice, they're sitting in their technology of choice through snowflake and through data sharing" [T1-01]. On scale: "We share data to hundreds of Snowflake accounts and we're managing hundreds of vendors, tens of thousands of pipelines", and "One of our biggest challenges is fleet management" [T1-01]. Pipelines are "moving away from batch oriented pipelines to more real time pipelines" using dynamic tables [T1-01]. Goel gives the latency: ADC is "a near real time data store ... which we think about as a data integration hub so that you can access all the data that you need in a lot in an e-front through one single source within 15 minutes of latency from various kinds of sources" [T2-050]. Jeff Miller, "Managing Director, Global Head of Data Platform and Product Engineering at BlackRock" and "a Chief Architect", describes what it replaced: "We sent flat files, we sent large schema definitions, FTP and related transfers, but the Aladdin Data Cloud and Snowflake makes it much simpler for us to get our clients' data and market data from our third-party providers and Aladdin partners back to our clients" [T1-17 Snowflake video, Jeff Miller, undated; Whisper].

Woodhead calls the content "the language of the portfolio", delivered as data products that are "designed, it's thoughtful, it's governed, it's documented, it has metrics around it" [T1-01]. Newman lists what is joining the network: eFront data, Preqin, "firms like London Stock Exchange data", and BlackRock IP "around credit analytics, around mortgage-backed security data, around insurance analytics", "all happening now" [T1-11]. Preqin indices "are also available today in Aladdin explore and our Aladdin data cloud capabilities" [T2-056 Unifying the Whole Portfolio, Sept 2025; Whisper].

The specs show the per-client structure. An ADC Dataset "maybe a current or past deployment of a table/view in Aladdin Data Cloud", with properties including `clientAbbreviation`, `database`, `schema`, `bitemporal`, `state`, `etlConfig` and `columnMetadata` [spec: AdcDatasetAPI]. Datasets move through draft, review and released states; release executes "DDLs in Snowflake which will create database objects such as table, views & stored procedure"; a reconcile operation compares a client's datasets "against another gold-copy client"; and the create, update, release and reconcile operations are each "intended for internal BlackRock use only" [spec: AdcDatasetAPI; briefing: suite architecture]. The API is "Used by Studio's ADC Admin Center" [spec: AdcDatasetAPI].

One client has spoken about the Snowflake part on BlackRock's channel. Hans Georgeson, CEO of Royal London Asset Management: "the implementation of Snowflake that we did as part of the overall programme has been really successful for us and we feel that it has moved us, it's leap-frogged us in many ways in terms of our ability to use data within the front office initially" [T2-058 Royal London client story, video file dated Nov 2024; Whisper]. He does not say "Data Cloud" or "Studio" [fact-check A1-6].

## 2.7 Developer tools

**AladdinSDK.** The README states the goal: "to make it easier for everyone to efficiently access and utilize Aladdin APIs and Data Cloud" [SDK README]. Installation is `pip install aladdinsdk` on Python 3.9, then

optional add-ons such as `asdk_plugin_trading` [SDK README]. The `AladdinAPI` client "wraps over OpenAPI generated python client code based on Aladdin Graph API's swagger specifications" and provides authentication, pydantic validation, response deserialisation, retries, "Rate limiting per API endpoint", error handling and "LRO Utilities to trigger and poll Long Running Operation endpoint calls" [SDK README]. Plugins are built by the `aladdinsdk-plugin-builder` project under github.com/blackrock using openapi-generator v6.6.0, triggered by git tags; the core package reads each plugin's `api_registry.py` and `domain_apis_list.json` to register its APIs, so new add-ons are picked up "without any additional setup steps" [briefing: Studio; SDK README; V8 S59].

Configuration "is performed using Dynaconf" with the environment prefix `ASDK_`, in increasing priority: default file, user file, `ASDK_` environment variables, inline parameters [SDK README]. `RUN_MODE` has three values: "AladdinSDK can run on a developer's local machine or in Aladdin Studio Compute sessions. There is also a test mode used by SDK developers" [SDK README]. In local mode the SDK "uses Keyring internally for secret management" and stores a prompted password in the OS credential manager [SDK README]. A log file per run, retries, pagination options, JSON-to-pandas conversion, Excel export and error emails "while executing their scripts / scheduled jobs" are built in [SDK README; V8 S41, S48, S49].

`ADCClient` "is a wrapper around snowflake-connector-python or snowflake-snowpark-python (connection type is configurable)" and can write a DataFrame to a table [SDK README; V8 S19]. `S3Client` wraps boto3 against "a Storage integration on Studio UI under a personal / shared project" [SDK README]. "DomainSDK Development" lets teams bundle the plugins they need into "business specific SDKs" on top of the core [SDK README].

**URL rewriting and gateways.** "When consuming the Aladdin APIs via an enterprise API management platform such as Azure API Management, the relative URL of the Aladdin APIs are likely to be different (in addition to the host name)", so the SDK accepts `API.URL_REWRITE.FIND` and `REPLACE` settings, a class argument or a per-call argument [SDK README]. The candidate's draft attributes the gateway URL rewriting and an additional-headers feature to two pull requests merged in December 2025 from a janushenderson.com address [REVISED draft Exhibit 33]; the fact-check lists this as not verifiable from the corpus, which holds the README and specs, not the repository history [fact-check A3].

**Studio names in the files.** The README and core specs name Studio UI, Studio Project (OAuth applications and storage sit "under a personal / shared project"), Studio Compute, Studio Notifications, S3 Storage Integration, API Applications, API Token, Studio Space and ADC Admin Center [SDK README; V8 Exhibit 4]. Studio Space appears only in notification settings, as a recipient entity and an event type, and is not defined in the files [V8 S4, S5]. BlackRock marketing calls the hosted environment Aladdin Compute, "a hosted analytics platform with ready-to-use Python environments and automation tools" [briefing: Studio]. The README refers to "'Aladdin Developer' or similar python development environments" where "a kernel restart is always recommended" after config changes [SDK README].

**Developer Portal and the rest.** The APIs page describes a portal with an API catalog, "API framework" documentation on how requests are built, authorised and processed, a Data Dictionary for browsing fields, downloadable Jupyter notebooks and multi-language code samples, all behind a client login; it says there are "dozens of APIs" with new ones added regularly, and it lists Aladdin Data eXchange (ADX) and Aladdin Tools for Excel (ATX) as Studio resources [briefing: Studio, citing the APIs page]. ADX "appears to be bulk data delivery for reconciliation and feeding other systems; ATX is an Excel add-in. Little public detail" [briefing: Studio].

## 2.8 Who builds on Studio

Two BlackRock speakers give the same internal figure. Woodhead: "inside of black rockck we have 3,000 citizen developers in business engineers that are interacting with all that in using Python or using R. I think that number has probably peaked. I think now we're on the other side of that where increasingly sophisticated workloads are going to be presented through natural language" [T1-01; auto-captions]. Goel: "every organization has a growing

number of citizen developers, or in fact, some people like to call them as embedded engineers now because they actually have a PhD degree in some cases ... When I reflected BlackRock, we have 3,000 people who qualify in this bracket today" [T2-050; Whisper]. Both figures count BlackRock staff, not Studio clients [fact-check A1-3].

Goel states the governance rule for that code: "when they are doing certain analysis, which are high risk, they have to promote these into production category. And at that point in time, they have to have certain governance around checking this code into GitHub in the same governance level that we check code from our software engineer as well as they need to have survivorship so that if they leave the organization, we can make sure that the code still runs after they have left" [T2-050]. He adds "a new class of employee, which I call as digital employees, or what we call as agents today" [T2-050]. Khara describes the user-provider model, with BlackRock using Aladdin "to manage the managed portfolios across the firm" [T2-061; verbatim in 2.10].

The Studio page names three personas (engineer, code-curious portfolio manager, data analyst), and the V8 document reads the files as serving two kinds of user: business staff who write Python (pandas and Excel export, settings that help "citizen developers share code-snippets or projects", the "break all citizen developers code" linter note) and IT teams that connect systems (enterprise API gateways, system account credentials, bot users) [page: Aladdin Studio via briefing: Studio; V8 conclusion 7; SDK README].

Outside firms named with Studio or Snowflake in the corpus: Royal London Asset Management, whose CEO credits the Snowflake implementation in its Aladdin programme [T2-058]; and Janus Henderson, through the SDK pull requests the candidate's draft reports from GitHub, unverified here [REVISED draft; fact-check A3]. On the Build vs Buy webinar, Clark Schafer of Alpha FMC says RFPs now ask "what's the API suite that you have? Are you using connectors to Snowflake or to data bricks?" and how quickly data moves, "every 15 seconds or ... every four hours?" [T2-049]. BlackRock has not named a client as building on Studio's APIs or the SDK; the SLC Management episodes placed on the Data Cloud page do not mention Data Cloud, Snowflake, Studio or APIs [fact-check A1-5, A1-6]. The draft's list of Barings, Thrivent, Franklin Templeton, Strata and EY as Studio users rests on job boards and third-party sites, not on the corpus [fact-check A3].

## 2.9 Copilot and Studio

Aladdin Copilot is "now available to all Aladdin clients" and "Will not give investment advice or respond to questions outside Aladdin platform boundaries" [page: Aladdin Copilot]. BlackRock's AI engineering lead Brennan Rosales and Principal AI Engineer Pedro Vicente Valdez describe a plug-in registry where "50, 60 Aladdin engineering teams" onboard in two ways: "we come to define a tool as kind of mapping one-to-one to an existing Aladdin API already out in production", or "the ability to spin up a custom agent" [T1-16 LangChain Interrupt 2025; Whisper]. Teams control "which environments this plug-in is enabled in, which user groups have access to this, and even which Aladdin apps can even access these plug-ins", and evaluation runs "on every PR" and "every day" [T1-16]. The Copilot tools are the same production APIs that Studio exposes; the packages themselves never mention AI, and the word "copilot" appears once, in a portfolio field about a distribution programme [V8 "What the files cannot tell us"; briefing: Studio].

Woodhead's AI strategy has three parts: AI in every product; "AI as a platform" for builders; and "how do I power the builder? So when I'm building on top of Aladdin and I need a co-pilot to help me build and solve the problem that I have, how does AI inside of the platform help me as an engineer, as a as an in business engineer, as a portfolio manager who's building or customizing or generating an artifact on top of the system?" [T1-01]. Newman's governance line: "speed to market doesn't necessarily imply better data. You still have to have the data governance. You still have to make sure that what is provided is correct"; and on AI, "If the data is wrong ... it might give you dangerous answers" [T1-11]. In the same episode he states the API roadmap: "we have these engines that historically were run through interfaces. So we're providing those APIs now. We're releasing soon the ability to actually access those analytic APIs, portfolio analysis, security analysis, things like that. That's all part of the Studio umbrella" [T1-11]. Goel adds that BlackRock wants "to start exposing some of the software and

application that we're building so that our clients can also access and leverage the AI that we developing" [T2-050]. Microsoft's Adrian Crockett, on BlackRock's podcast: "There's no copilot without plugins" [T2-023 Microsoft Copilot Episode Part 2, 6 Nov 2024; Whisper]. Pavan Pemmaraju, Head of AI Product & Engineering, says "the data model itself is structured for agentic use cases" [T2-028 Reducing time to insight, 16 Jul 2026; Whisper, via fact-check A5-11].

## 2.10 How BlackRock presents Studio

What is public: the Studio page, whose calls to action are "Book a meeting" and "Download the brochure"; the Studio for Aladdin Risk page; the APIs page with "dozens of APIs"; the Data Cloud page; the SDK README, GitHub repositories and PyPI packages; the engineering blog post on the SDK; the Woodhead and Newman podcasts; the Fivetran write-up; the Jeff Miller video; and the LangChain talk [briefing: Studio; REVISED draft Exhibit 29; T1-01; T1-11; T1-16; T1-17]. What is gated: the API catalog, API-framework documentation, Data Dictionary, notebooks, code samples, rate limits, SLAs, pricing and the Studio UI [briefing: Studio §8]. The SDK needs the client's Aladdin host URL and a token or OAuth application "registered via Studio UI" before any call succeeds [SDK README; fact-check A4].

Job postings on BlackRock's careers site, as recorded in the briefing and the candidate's draft (seen 8 October 2026): a Senior Product Director, Aladdin Studio Developer, describing Studio Developer as "Aladdin's extensibility layer" that lets clients "build, adapt, and run custom workflows in a governed, enterprise-ready way", with goals to "reduce off-platform development risk" and "translate internal success into client adoption"; a Technical Writer, Aladdin Studio, covering "Open APIs and event-driven workflows", AsyncAPI, and an "Aladdin Studio Developer Portal"; a Studio Developer Operations Vice President asking for Python or Go plus Kubernetes/AKS; a Forward Deploy Engineer, Aladdin Data, in Mumbai, naming "Model Context Protocol (MCP), AI agents, and Copilot extensions"; and a Studio product-marketing posting live by November 2024 saying Studio has "free and premium capabilities" [briefing: Studio §1, §6, §7; REVISED draft Exhibits 14 and 28]. The posting texts are not in the pulls.

The leaders who speak for Studio, with titles as the corpus gives them: Dave Woodhead, "managing director and head of engineering for Aladdin Studio" [T1-01] and "BlackRock's Head of Aladdin Studio Engineering" [page: Fivetran blog]; Richard Newman, "General Manager, Aladdin Studio" [T1-11, page header]; Diwakar Goel, "Global Head of Data" for Aladdin [T2-050]; Jeff Miller, "Managing Director, Global Head of Data Platform and Product Engineering" and "Chief Architect" [T1-17]; Kunal Khara, "Global Head of Aladdin Product" [T2-010; T2-061]; Pavan Pemmaraju, "Head of AI Product & Engineering" [T2-028]; Kamya Somasundaram, "Global Head of Aladdin Partnerships & Trading Product Marketing" [T2-017]; Lance Braunstein, "Managing Director, Aladdin" [T2-022]. Khara's two Studio lines, verbatim: "What we call the Aladdin studio estate, we think is going to this year be a major, major unlock, because we're finding that more users are craving this ability to interact with our data, with their data, with reporting with dashboards on a programmatic basis at machine scale, over and above what they get through our front end applications as well" and "Ultimately, we eat our own cooking at BlackRock" [T2-061 Best Overall Technology Provider 2023; Whisper].

The same interview lists Khara's 2024 themes: generative AI on a data foundation, cloud elasticity, the network ecosystem, the whole portfolio, and Studio [T2-061]. The Big Little Questions series frames the audience as a client trend: "Now they want programmatic access to the data to build various kinds of applications. And that's a huge shift that has happened in the last five to seven years" [T1-26 BLQ: In the age of AI, what is data today].

## 2.11 What the pulls do not show

- No public event specification. The 54 specs are OpenAPI request-response definitions; no AsyncAPI document, webhook or stream specification is in the packages. Events exist in BlackRock's description

(Woodhead, Goel, Sewdass) and in job-posting summaries only [digest; T1-01; T2-050; T2-049; briefing: Studio].

- No pricing or packaging. No Studio price, SLA, rate limit or tier appears in any BlackRock-published source in the corpus. The one packaging phrase, "free and premium capabilities", comes from a job posting recorded in the candidate's draft [REVISED draft Exhibit 14; briefing: Studio §8].
- No named external SDK or API client. Royal London credits Snowflake; no client on a BlackRock channel describes building on the APIs or SDK [T2-058; fact-check A1-6]. The Janus Henderson pull requests are reported from GitHub, outside the corpus [fact-check A3].
- Core-package API counts. The core `aladdinsdk` package's four APIs (TokenAPI, Studio Subscription, Studio Notification, a TrainJourney demo) and 25 operations, and the totals that include them (58 APIs, 326 operations), are the candidate's own tally; only the 11 add-ons' 54 specs and 301 operations were recomputed and verified [V8 "What the packages contain"; fact-check B3].
- The full catalog. BlackRock says "dozens of APIs" and Newman says analytics APIs were "releasing soon" in January 2025; positions and holdings, risk analytics outputs, performance attribution, pricing and market data, benchmarks, execution and FIX connectivity, settlement messaging, eFront and wealth APIs are absent from the public packages [briefing: suite architecture §2; T1-11].
- The Studio screens, Studio Space, Compute scheduling and the ADC Admin Center UI: named in the files, not described; no scheduler inside Studio is shown [V8 Exhibit 4 and conclusion 6].
- Usage. No BlackRock source counts Studio clients, API calls, SDK installs or Compute sessions; the 3,000 figure is internal headcount [T1-01; T2-050].
- The meaning of `Access.Internal`, the expansion of the ACM event type, and the dates APIs moved between alpha, beta and stable [briefing: suite architecture §5; V8 verification note 3]. Release history (first release May 2024; add-on releases 17 July 2024 and 22 January 2026) is the candidate's reading of PyPI; only 2.0.0b10 in July 2026 was verified [V8 Exhibit 12].

# Part 3. BlackRock

## 3.1 The Aladdin business inside BlackRock

BlackRock is the first user of Aladdin and the vendor of it. Brennan Rosales put it as "BlackRock is the biggest user of Aladdin, but we also sell and support hundreds of clients globally across 70 countries all using the Aladdin platform" [T1-16 LangChain Interrupt, 2025]. Dave Woodhead gave the origin in the same terms: "As BlackRock grew, our clients recognized that our technology was differentiated and asked could they use the same software. So BlackRock pretty quickly entered the fintech business and we started offering Aladdin as a service" [T1-01 Snowflake Data Cloud podcast, 2024]. James Sword described "BlackRock Aladdin, the business unit within BlackRock that builds our portfolio management software Aladdin, used by BlackRock and its clients alike to manage their whole portfolio solutions across public and private assets" [T2-052 Navigating private markets webinar, undated].

The size of the organisation is stated once in the pulls: "The Aladdin organization is made up of around 7,000 people, with over 4,000 of those being engineers", who "build and maintain the around 100 Aladdin front-end applications that thousands of people use every day", organised in "say, 50, 60 Aladdin engineering teams that cover specific domains throughout the platform" [T1-16 LangChain Interrupt, 2025]. The ratio of internal to external use is not given; the only internal headcount figure is the 3,000 "citizen developers and business engineers" inside BlackRock, stated by Woodhead and again by Diwakar Goel ("we have 3,000 people who qualify in this bracket today") [T1-01 Snowflake Data Cloud podcast, 2024; T2-050 Data as a strategic asset for pensions, 2025].

"We eat our own cooking" is Kunal Khara's phrase: "we are a user provider at the core. Ultimately, we eat our own cooking at BlackRock. We use Aladdin to manage the managed portfolios across the firm" [T2-061 BST Awards winner interview, 2023 awards]. Sue Zheng put the engineering consequence: "in Aladdin we always try to push the same software out to all clients ... I don't want to be on the hook for saying I have like 50 different versions out there", and "it's also important for the client to know that BlackRock, the house, is using the same software, right? We have skin in the game" [T2-027 Principles of Design, May 2023]. Afwa Kandawire added that "Every single client environment looks the same, is built the same way, we have a single control plane and a single source of truth", and "That notion of a single code base at all clients leads to simplicity, it's what enables the scale, it's what enables the reliability" [T2-027 Principles of Design, May 2023]. Lance Braunstein calls the same thing "the user provider model of BlackRock, the world's largest asset manager" [T2-037 The Way Forward, May 2024].

Internal use is described concretely. BlackRock's private equity partners business now wires capital calls through Aladdin's SWIFT capability, "moving from something that took 24 hours now down to one hour" [T2-056 Unifying the Whole Portfolio webinar, Sept 2025]. The private credit data model was designed "from a user provider perspective with our professionals at BlackRock" and then with "our new colleagues from HPS" [T2-056]. Darren Cannon is introduced as the person who runs "BlackRock's private market operations from investor onboarding, middle office and now production" on Aladdin and eFront [T2-052]. The accounting team says "we're the user of the same software ... the way we use it is the way our clients use it" [T1-33 Big Little Questions, accounting episode].

Teams named by BlackRock speakers in the pulls: the Aladdin Client Business (Tarek Chouman; Pia Schaefer's Value Engineering sits inside it), Aladdin Engineering (Braunstein), Aladdin Studio engineering (Woodhead) and Aladdin Studio as a product with a General Manager (Richard Newman), Aladdin Product (Khara), Aladdin Wealth Tech (Ted Stratigos; James Werner for APAC), Data Platform and Product Engineering (Jeff Miller), a Global Head of Data (Goel), AI Product and Engineering (Pavan Pemmaraju), Product for AI (Syril Smith Garson), Aladdin

Platforms (Kandawire, with product families she lists as mission control, hosting services, engineering services, enterprise architecture and a product office), Solutions Engineering (Cameron Skinner globally; Alisha Sewdass for the Americas; Sword for private markets; Sanaya Khaneja), Aladdin Partnerships and Trading Product Marketing (Kamya Somasundaram), Aladdin Business Development (Tim Garbien for the Americas; Hui Min Loh), Aladdin Client Engagement (Tara Kodali, James Davidson, Alton Garrity for pensions), Aladdin product marketing (Said Shah, Nick Codge), Product Marketing and Climate Solutions (Tracy Weldon), the Aladdin sustainability team (Matt Guinness), the Aladdin private markets product team (Paul McGaugh) and RQA, BlackRock's risk management group (Rory Van Swannenberg) [sources as in the table in 3.2].

No BlackRock speaker or page in the transcripts and pages states a revenue, ACV or client-count figure for the technology business. The Studio briefing records, from BlackRock job postings and quarterly results outside the transcript set, that postings call Aladdin a "$1BN+ technology business" and that Q2 2026 technology services revenue rose 13% with ACV up 15% [briefing: Studio]. Two figures for a client segment do appear: Georgette Anderson said "we have roughly about 500 billion of insurance asset management on the system and 6 trillion of insurance assets that are actually processed through Aladdin" [T2-040 Big Little Questions, insurers].

## 3.2 People who speak for Aladdin in the pulls

| Name | Title as the corpus gives it | Where | Responsible for, as stated |
|---|---|---|---|
| Sudhir Nair | "Global Head of Aladdin"; "Global Head of the Aladdin Business" (2023) | T2-033 Art of the possible, Jun 2025; T2-005 Columbia Threadneedle, Jul 2024; T2-006 AustralianSuper, May 2023 | Runs the Aladdin business; 25 years at BlackRock from the graduate analyst programme; "leads a really vibrant technology organization" [T2-033] |
| Dave Woodhead | "managing director and head of engineering for Aladdin Studio" (Snowflake podcast); "Head of Aladdin Studio Engineering" (Fivetran blog) | T1-01 Snowflake Data Cloud podcast, 2024; page: Fivetran blog, Aug 2024 | Engineering for Studio: APIs and events, Aladdin Data Cloud on Snowflake, developer tools [T1-01] |
| Richard Newman | "General Manager, Aladdin Studio" | T1-11 Never Done: Data, the key ingredient (also T2-013, Jan 2025) | "my role here at Aladdin is to be very much between Commercials and Engineering" [T1-11] |
| Diwakar Goel | "Global Head of Data from Aladdin" (Whisper renders him "the boxer, Guell") | T2-050 Data as a strategic asset for pensions, 2025 | Data and AI platform strategy; opening Aladdin through events, APIs and ADC [T2-050] |
| Jeff Miller | "Managing Director, Global Head of Data Platform and Product Engineering at BlackRock ... Chief Architect and a member of our Enterprise Architecture team" | T1-17 Snowflake video, undated | Aladdin Data Cloud; replaced "flat files ... FTP" distribution [T1-17] |
| Kunal Khara | "MD and Global Head of Aladdin Product" | T2-061 BST Awards interview; T2-010 Global Insurance Report, Nov 2023; T2-037/038 The Way Forward, 2024; T1-30 Big Little Questions | Product roadmap; named Studio as a 2024 focus [T2-061] |
| Pia Schaefer | "Global Head of Value Engineering for the Aladdin Client Business" | T2-011 Change alpha with McKinsey, Jul 2025; T2-012, Apr 2024 | "change alpha", operating leverage from the platform [T2-011] |
| Ted Stratigos | "Global Head of Aladdin Wealth Tech" | page: Aladdin Wealth Auto Commentary press release, 2 Oct 2025 | Aladdin Wealth; Auto Commentary launch with Morgan Stanley |

| Name | Title as the corpus gives it | Where | Responsible for, as stated |
|---|---|---|---|
| Cameron Skinner | "Global Head of Solutions Engineering for the Aladdin business"; "Managing Director and Head of Aladdin Solutions Engineering at BlackRock" | T2-016 Citisoft, Jan 2026; T2-056 Whole Portfolio webinar, Sept 2025; T2-041 Big Little Questions; T2-045 FT FundFire webinar | Solutions design for prospects; "a decade here at BlackRock" [T2-016] |
| Lance Braunstein | "Managing Director, Aladdin" (2023); "head of Aladdin Engineering at BlackRock" (Microsoft, Dec 2024) | T2-022 Microsoft partner story, Mar 2023; T2-037/038, 2024; page: Microsoft Cloud blog | Engineering; cloud and AI journey with Microsoft |
| Georgette Anderson | no title given | T2-040 Big Little Questions, insurers | Insurance segment; gives the $500bn / $6tn figures |
| Pavan Pemmaraju | "Head of AI Product & Engineering" | T2-028 Reducing time to insight, Jul 2026 | "leading our AI strategy" per the host; agents, memory, guardrails |
| Jess Fitzgibbon | "Head of Product for Aladdin Risk"; "manager director and head of product at Aladdin risk" | T2-028; T2-051 FT FundFire webinar | Aladdin Risk product |
| Kamya Somasundaram | "Global Head of Aladdin Partnerships & Trading Product Marketing" | T2-017 Aviva Investors, Dec 2025 | Trading network (FIX, "hundreds of participants"), partnerships |
| Brennan Rosales; Pedro Vicente Valdez | "AI engineering lead at BlackRock"; "Principal AI Engineer" | T1-16 LangChain Interrupt, 2025 | Aladdin Copilot orchestration, plug-in registry, evaluation |
| Afwa Kandawire | "Global Head of Aladdin Platforms" | T2-027 Principles of Design, May 2023 | Platform engineering: uptime, hosting, developer friction, enterprise architecture |
| Sue Zheng | "Tech Fellow"; "leads Global Engineering for the whole portfolio solution and private credit" | T2-027; T2-052 | Multi-asset data model, private credit |
| Syril Smith Garson | "Head of Product for AI" (2025); "Head of Product for Alpha, Research and AI platforms" (2023) | T1-12/13 Never Done with Microsoft; T2-019, Nov 2025; T2-035, Sept 2023; T1-27 | Copilot platform, "common services and capabilities that underpin all of these different copilot experiences" [T1-13] |
| Adam Salvatori | "Head of AI, Alpha, and Digital Assets Engineering" | T2-035 Fourth industrial revolution, Sept 2023 | AI and digital assets engineering |
| Tarek Chouman | "Global Head of the Aladdin Client Business"; former CEO of eFront | T2-014, Sept 2025; T2-030, Dec 2024; T2-058, T2-059, T2-063 | Client business; Provider network; private markets |
| Mark O'Hare | "Vice Chairman for BlackRock"; founder of Preqin | T2-014 Democratizing private markets, Sept 2025 | Preqin inside BlackRock |
| Tim Garbien | "Head of Aladdin Business Development for the Americas"; "managing director of Aladdin at BlackRock" | T2-021 Alpha FMC, Mar 2026; T1-18 Summit for Asset Management | "travel the earth talking to asset managers and asset owners" [T1-18] |
| Woo Fung Kwong | "Managing Director, Chief Strategic Advisor to Aladdin" | T2-036 Morgan Stanley, recorded Q3 2023 | Aladdin Wealth origins |
| Robert Mitchnick | "Head of Digital Assets, BlackRock" | T2-015 Coinbase, Apr 2026 | Digital assets; partnership strategy |
| James Werner | "head of Aladdin WealthTech APAC" (Whisper: "London WealthTech APAC") | T2-039, Mar 2026 | APAC wealth |

| Name | Title as the corpus gives it | Where | Responsible for, as stated |
|---|---|---|---|
| Melissa Faraz; Paul McGaugh | "global head of business development for private markets, analytics platforms and head of Preqin London"; "managing director in our Aladdin private markets product team" | T2-056, Sept 2025 | Preqin integration; whole portfolio and private credit |
| Alisha Sewdass; Michael Campos | "runs our Aladdin Solutions Engineering team for Aladdin in the Americas"; "I cover the US Asset Managers for Aladdin" | T2-049 Build vs Buy webinar | Build vs buy framing |
| Randy Fradin; Lou Rosato; Marija Djordjevic; "Shirley Tvett" | no titles in the files (Rosato speaks for BlackRock operations; Tvett is "Managing Director and Chief Operating Officer at Aladdin BlackRock" as Whisper renders it) | T1-28; T1-10 Symphony Innovate 2026; T2-043/044 Uncharted; T2-046 FT webinar | Azure migration; interoperability; private markets benchmarking; FT panel |
| Microsoft counterparts | Adrian Crockett, "General Manager, Microsoft Cloud for Financial Services"; Jason Wild, "Vice President, CEO Co-Innovation and Customer Engagement"; Matthew Kerner; Allison Nachtigal; Judson Althoff, "Executive Vice President and Chief Commercial Officer" | T2-019/023, T1-12/13; T2-022; T1-28; T1-27; page: Microsoft Cloud blog | Copilot design; partnership; Azure migration; Ignite 2024 |
| Snowflake, Fivetran counterparts | Christopher Napoli, "Head of Wealth & Asset Management, Snowflake"; Taylor Brown, COO, Fivetran | T1-11, T2-013, T2-050; page: Fivetran blog | Data Cloud partnership; pipelines |
| NVIDIA counterpart | none; the GTC page was JavaScript-only and holds no names | page: NVIDIA GTC (empty) | not retrievable |

## 3.3 Strategy as stated

**Whole portfolio and a common language.** Khara: "our entire focus since that time that even today has been about supporting the investment lifecycle for the whole portfolio the whole way through ... Aladdin unifies the entire investment management process through a common data language" [T2-061 BST Awards interview]. Woodhead: "we bring all of that together into a single symbology, a single language, the language of the portfolio" [T1-01]. The 2026 Client Forum opener: "So we built a common language for portfolios. Today, we're building on it, an investment intelligence platform powered by AI-native intelligence" [T2-060 Aladdin Client Forum 2026]. Melissa Faraz: "while Aladdin lives as is the common language for portfolios. We really want a lot and inclusive of a front and prequin to also be the common language of data" [T2-056, Sept 2025]. Chouman: "BlackRock and Aladdin was built for the concept of the whole portfolio" [T2-063 APAC Client Forum recap, 2025], and "The whole portfolio today, we are speaking about it, we believe that in few years it will become the norm" [T2-014, Sept 2025]. Woodhead: "we are very much on a journey to what we call a whole portfolio view ... We don't think anybody else will be able to provide that" [T1-01]. The product page claims "6 years delivering Whole Portfolio solutions—with 20+ years delivering solutions for private markets" and "50,000 unique private markets companies maintained and 2k+ private markets risk factors monitored daily" [page: Whole Portfolio].

**Public and private together.** Nair: "Aladdin grew up very much as a public markets oriented technology. And it wasn't until 2019, when we did our first major technology acquisition with E-Front ... The private market side was still very opaque ... which led us to about three months ago, we closed our next acquisition, which is a company called Pregwin" [T2-033, Jun 2025]. Newman spoke of "the announcement of the Preqin acquisition" and "eFront's an obvious example of that, that data coming into the Aladdin Data Cloud" [T1-11]. The acquisition video welcomes "the 1600 colleagues from prequin to BlackRock" [T2-062]. O'Hare: private markets were "4 or 500 billion of dollars of AUM worldwide" in 2002, "Today, it's $13 trillion", with a Preqin mantra of "30 by 30" [T2-014]. Goel: "we actually want to make it available across the whole Aladdin Suite, which is core Aladdin e-front as well

as Aladdin wealth. We want to make data like the fund data, the benchmark data also available in our analytical platforms like the Explore and ADC" [T2-050]. Faraz, September 2025: "investors now have access to over 15,000 fund investments within e front insight"; "Preqin indices are also available today in Aladdin explore and our Aladdin data cloud"; indices on BlackRock's strategic asset allocation tools and on Aladdin Wealth were "working to embed" and "looking to integrate" [T2-056]. McGaugh described "over 10 different transaction types" added for loans and a private credit IBOR; Faraz said it launched "in less than a year post acquisition" [T2-056]. The Studio briefing dates Preqin's fold into eFront Insight to February 2026 [briefing: Studio]. A May 2026 private credit analytics release is asserted only in the candidate's revised draft and is not in the pulled transcripts or pages.

**Cloud.** Microsoft's Ignite recap: "Five years ago, we began working on a strategy to move them into the cloud on Microsoft Azure. Today, all BlackRock's technology platform clients and services are hosted in the Azure cloud ... New rollouts now take weeks instead of quarters to complete" [page: Microsoft Cloud blog, 11 Dec 2024]. Kandawire: "the first stage would be a lift and shift ... the principal reason for doing a lift and shift was to ensure that we could actually enable Aladdin to be available to more prospects in more regions than we had datacentres available" [T2-027, May 2023]. From the Fradin and Kerner episode, whose Whisper file does not label speakers: "Initial step of moving the Latin into Azure was always meant to be the springboard to what comes next, which is essentially a perpetual cloud journey" [T1-28]. Braunstein, on why Microsoft: "you're great at data centers. We will never be the data center operator that you are, and that's fine" [T2-022, Mar 2023]. The AWS option announced December 2025 with general availability in H2 2026 appears in the briefings, not in any transcript or page [briefing: suite architecture].

**Openness.** Khara listed "this big focus on open architecture and access to APIs" as "in very high demand over the last few years", and for 2024: "What we call the Aladdin studio estate, we think is going to this year be a major, major unlock, because we're finding that more users are craving this ability to interact with our data, with their data, with reporting with dashboards on a programmatic basis at machine scale" [T2-061]. Goel: "we really want to make every single event that happens in a lot in available to you. So every single transaction that happens should be available and accessible to you via an event, via an API"; ADC as "a data integration hub ... within 15 minutes of latency"; and "extending a lot in into a platform", with investment automation as the first use case [T2-050]. Woodhead's three pillars: "It exposes everything Aladdin knows how to do as a set of APIs events. The Aladdin data cloud built on top of Snowflake ... And increasingly we're building a set of developer tools" [T1-01]. Newman: "The vision's always been other firms join that ecosystem. So, you know, bringing in, like, firms like London Stock Exchange data into the ecosystem" [T1-11]. Chouman on the Provider network: "we've expanded our provider network to include nine of the largest global asset services" [T2-059]. Somasundaram: "there is a burgeoning community of individuals saying 'I need that in the form of an API. I need that to speak to this.'" [T2-017]. Chouman in APAC: "our strategy is to meet where you are" [T2-063].

**AI.** Khara, early 2024: "on the E-front side in particular, we have a co-pilot that is out in the market very soon, there'll be something we're going to be doing on the Aladdin front as well" [T2-061]. The FT webinar panellist for Aladdin: "we recently introduced the Aladdin Co-Pilot ecosystem with a select number of clients. The first example of that was e-front co-pilot" [T2-046]. By December 2024 Microsoft wrote "BlackRock technology clients are already using this capability today" [page: Microsoft Cloud blog]; the product page says "now available to all Aladdin clients" [page: Aladdin Copilot]; by 2025 Copilot was "embedded across all of those 100 front-end applications" [T1-16]. Microsoft also records BlackRock "starting an AI Lab in 2018" [page: Microsoft Cloud blog]. Woodhead's "AI strategy in in my mind is really three-fold": AI "woven into everything we do", "AI as a platform" for builders, and "a co-pilot to help me build" [T1-01]. Pemmaraju and Fitzgibbon, July 2026: "We're not at a point of the agents making decisions. It's human in the loop", and Pemmaraju: "I see a 12 to 18 month time frame. I think I definitely see high confidence, low risk actions can potentially be automated" [T2-028]. Goel: "a new class of employee, which I call as digital employees, or what we call as agents today" [T2-050]. Chouman: "a futuristic view of trading. One where a portfolio manager is going to be able to talk to Aladdin, where agents are going to be able to work together to make decisions" [T2-063].

**Direction, verbatim.** Khara, May 2024: "our goal is to first, meet them where they are, but ultimately to take them where they need to go. So, it's not just about how we adapt to change, but it's actually how we drive the change as well" [T2-037]. Nair, June 2025, on what excites him over two to three years: "this idea of private markets ... I think it will happen in a much shorter time frame than 25 years" [T2-033]. Woodhead: "We want to live in a world where any user can with natural language ask any sophistication of question against our entire data corpus and we can with explanability answer that through AI" [T1-01]. Goel: "we need to have one single data and AI platform for the firm" [T2-050]. Pemmaraju: "portfolio becomes a living system" [T2-028]. Braunstein, June 2024: "blockchain, and I'm not talking about crypto ... I mean, a token ... to me is foundational" [T2-038].

## 3.4 Partnerships named in the pulls

| Partner | Role as the corpus states it | Source |
|---|---|---|
| Microsoft | Azure hosting of all Aladdin clients; Copilot co-development ("A lot of the new AI development taking place now is a close engineering partnership between the BlackRock and the Microsoft team", Fradin and Kerner episode, speaker not labelled); Azure OpenAI and Azure ML; M365 and GitHub Copilot, Copilot Studio and Dynamics 365 used internally; Ignite 2024 stage. A Nadella/Fink 2020 announcement is not in the corpus. | T1-28; T2-022; T1-12/13; T2-019/023; T1-27; page: Microsoft Cloud blog |
| Snowflake | Platform for Aladdin Data Cloud; "hundreds of Snowflake accounts"; Data Cloud podcast; Snowflake Summit 2024 talk; audiocast with Napoli; pensions webinar | T1-01; T1-11; T1-17; T2-050; page: Fivetran blog |
| Fivetran | Data movement into ADC's Snowflake landing zone; "wasn't possible without Fivetran" (Woodhead) | page: Fivetran blog, 28 Aug 2024 |
| LangChain | Copilot orchestration "built using Langchane" from day one; LangSmith traces; GPT-4 function calling; evaluating agent protocols | T1-16 |
| NVIDIA | A GTC session (S72854) exists in the link directory, but the page was JavaScript-only and nothing was retrieved | T1-00 index; page: NVIDIA GTC |
| Amazon Web Services | Hosting option announced December 2025, GA H2 2026; not in any transcript | briefing: suite architecture |
| HSBC | Third partner on Royal London's implementation: "yourselves in Aladdin, ourselves in Royal London, and also, of course, HSBC as our partners" (Georgeson) | T2-058 |
| J.P. Morgan Securities Services and eight other asset servicers | Aladdin Provider network; JPMorgan "one of the very early adopters ... from 2016"; "nine of the largest global asset services" | T2-059 |
| BNY | Named with BlackRock in AIA's July 2024 agreement | page: Whole Portfolio |
| iCapital | Aladdin Wealth proposal to subscription workflow, "the first time we've done something like this" | T1-08 iCapital Connect |
| Symphony | Interoperability "bridge" for BlackRock operations; Enhanced Directory in Aladdin | T1-10 Symphony Innovate 2026 |
| Coinbase | "Coinbase was the first of those" partnerships for BlackRock digital assets (Mitchnick); not an Aladdin connectivity claim | T2-015, Apr 2026 |
| London Stock Exchange | Named by Newman as the kind of data firm the ecosystem vision brings in | T1-11 |
| Citisoft; Alpha FMC; McKinsey | Consultancies co-hosting podcasts and the Build vs Buy webinar | T2-016/029; T2-021, T2-049; T2-011 |
| Financial Times / FundFire; ILPA | Webinar series sponsor and co-host; eFront FAIR session with ILPA | T2-045/046/051; T2-054 |

| Partner | Role as the corpus states it | Source |
|---|---|---|
| Preqin, eFront, HPS, GIP | Acquisitions, "three significant private market deals ... in the past year, alongside global infrastructure partners and HPS" | T2-056; T2-033; T2-014 |

## 3.5 Clients BlackRock features

| Client | Segment | What was said | Source |
|---|---|---|---|
| Columbia Threadneedle | Asset manager | CEO Ted Truscott with Nair; "in the research department, we won't hire anybody who can't program in R or Python" | T2-005, Jul 2024 |
| Intesa Sanpaolo | Asset manager / bank | Paola Papanicolaou, Group Head of Transformation Area; "we decided together with you to do a full adoption" | T2-002, Mar 2024; T2-020 |
| BNP Paribas Asset Management | Asset manager | CTO Fatia-Fatma Balit: "Back in the Race" 2016 to 2020 "with BlackRock Aladdin in the front to back chain, with Salesforce in the CRM, and transversally DataHub"; "decommission more than 60 applications" | T2-008, Feb 2023; T2-009 |
| Royal London Asset Management | Asset manager | CEO Hans Georgeson: "largely on time, if not ahead of plan, and we came in under budget"; one order management system replacing "multiple"; "the implementation of Snowflake that we did as part of the overall programme has been really successful" | T2-058, recorded Nov 2024 |
| TCW | Asset manager | Global COO Melissa Stolfi on AI and scale | T2-018, Jul 2026 |
| Aviva Investors | Asset manager | Global Head of Trading Kate Hill: "we've been using rules-based execution, I think now coming up a decade" | T2-017, Dec 2025 |
| Franklin Templeton, Russell Investments, Barings, AXA IM ("Accident Investment Managers") and a Swiss group CTO | Asset managers | Webinar panellists alongside Aladdin speakers; no adoption statement | T2-045, T2-051, T2-046 |
| SLC Management | Insurer-owned asset manager | Candace Shaw, Deputy CIO and COO, on data practices and legacy systems | T2-025, Mar 2026; T2-031, Feb 2024; T2-032 |
| Helvetia | Insurer | CTO Dominik Gurtner; the audio transcript is garbled, the page description says "holistic, real-time view across public and private assets" | T2-057; page: Whole Portfolio |
| AIA | Insurer | "AIA announces a new agreement with BlackRock and BNY as it transforms its investment platform" (12 July 2024) | page: Whole Portfolio |
| AustralianSuper | Pension | COO Peter Curtis with Nair on digital transformation | T2-006, May 2023; T2-007 |
| University Pension Plan (UPP) | Pension | CIO Aaron Bennett on trusted data and the total portfolio approach | T2-034, Sept 2026 |
| Ingka Investments | Corporate investor | Samuel Rundle: "critical for us was to create more transparency"; moved to electronic execution; look-through on private funds | T2-030, Dec 2024 |
| Morgan Stanley Wealth Management | Wealth | Chris Scott-Hansen on risk-centred advice; first to implement Auto Commentary, October 2025 | T2-036, recorded Q3 2023; page: Wealth Auto Commentary |
| J.P. Morgan Securities Services | Asset servicer | Naveen TV, Global Head, Data strategy and Integrations; Provider early adopter; reconciliation-break workflow live | T2-059 |

| Client | Segment | What was said | Source |
|---|---|---|---|
| Northstar Group | Private equity (APAC) | Head of ESG Priya Gopalan on an ESG toolkit implemented in 2023 | T2-003, May 2024; T2-026 |
| Ardian | Private markets | Teaser on automated investor-portal data | T2-004, Aug 2023 |
| eFront testimonial videos | Private markets GPs, LPs, servicers | 6M, Adepa, Allianz Capital Partners, Alter Domus, Andera Partners, APG Asset Management, BlackRock Alternative Investors, BNP Paribas Securities Services, Bouwinvest, Brown Brothers Harriman, CapMan, Dunedin, EDC, IPS Fund Services, Madrona Venture Group, MB Fund, Meridiam, Mirova, Mizuho, Neo Capital, Ocorian, Old Mutual Alternative Investments, Social Investment Scotland, Vaekstfonden, Velliv | T2-064 to T2-076, T2-084 to T2-095 |
| Microsoft Treasury | Corporate | First environment moved to Azure, per the briefing only | briefing: Studio |

## 3.6 How BlackRock talks to the market

The channels found are podcasts, a video series, webinars, client-story videos, event videos, partner stages and one press release.

**Podcasts.** "Aladdin Client Stories" runs from February 2023 (BNP Paribas AM) and "Never Done by Aladdin" from a trailer dated 28 May 2025, hosted by Meera Jessa, whose intro is "Today, technology is truly never done" [T2-00 index; T2-024; T2-033]. Episodes pair a BlackRock lead with a client, consultant or partner. Engineer-led episodes exist: "Principles of Design" with Kandawire and Zheng [T2-027], and "The fourth industrial revolution" with Smith and Salvatori [T2-035].

**Big Little Questions.** "A new series by Aladdin, delivered by the people on the front lines of change" [page: Big Little Questions]. Episode questions on the page: How can tech transform numbers into insightful narratives? What is the potential of AI in financial services? How is emerging tech reshaping the user experience? How do clients shape product development? How are analytics reshaping how we manage risk? As tech evolves, how do you create better client solutions? What is the role of emerging tech in financial services? In the age of AI, what is data today? In fintech, where does the platform end and the solution begin? In financial services, how have relationships evolved? Where's the light at the end of the tunnel for pensions? Is ecosystem an empty word? How do private markets players define risk anymore? Are insurers disruptive or disrupted? What does scale mean for asset managers? [page: Big Little Questions]. Transcripts cover twelve of these plus an asset-managers episode whose opening line is "is efficiency a dirty word" [T1-21 to T1-33; T2-040 to T2-042].

**Build vs Buy webinar.** Hosted by Michael Campos with Sewdass and Alpha FMC's Clark Schafer. Poll one: "mostly built in house at about 15%. The next is actively reevaluating at 18%, hybrid second most at 30 and then mostly vendor platforms at 37%". Cost slide: "6070% of tech costs or maintenance of current systems", which Schafer restated as "60 to 70% of your, your tech organisation focused on keeping the lights on". Poll two: "AI and data demands at 42% an exact tie between cost and speed to market And last place was regulatory complexity". Poll three: "compliance reporting is number one at 53% followed by position and cash management 22%, trade life cycle at 17 and finally risk infrastructure at 9%". Timeline: "roughly 12 to 18 months", with ROI "about that six month post implementation". The RFP line: "what's the API suite that you have? Are you using connectors to Snowflake or to data bricks?" Closing: "maintaining plumbing or functions that are solved or commoditized may not be the best source of generating alpha, generating differentiation" [T2-049].

**Events.** Aladdin Client Forum 2026 in New York, with a produced opener ("This is Aladdin, built by you, the intelligence of investing") [T2-060]; an APAC Client Forum recap for 2025 [T2-063]; cadence not stated. Snowflake Summit 2024 (Woodhead with Fivetran) [page: Fivetran blog]; Snowflake's Data Cloud podcast [T1-

01]; LangChain Interrupt 2025 [T1-16]; Microsoft Ignite, Chicago, November 2024, written up by Althoff [page: Microsoft Cloud blog]; Symphony Innovate 2026 [T1-10]; iCapital Connect [T1-08]; the Summit for Asset Management East Coast [T1-18]. NVIDIA GTC: a session exists but was not retrievable [T1-00 index].

**Awards.** "BlackRock won three categories in this year's BST Awards, Best iBOR or Investment Book and Record platform, Best Portfolio Analysis Tool and Best Overall Technology Provider for 2023", in WatersTechnology's winner interview with Khara [T2-061]. Royal London and Aladdin also "got an award, jointly" for the implementation, unnamed [T2-058].

**Other formats.** The Uncharted series on private markets benchmarking with Marija Djordjevic [T2-043/044]; webinar replays behind registration, including FT FundFire series, Aladdin Climate, trade tensions with Market Driven Scenarios in Explore, pensions data, private markets, the whole portfolio, and eFront sessions [T2-045 to T2-056]; client-story videos [T2-057 to T2-059]; the Global Insurance Report podcast [T2-010]; "The Way Forward" white paper and two podcasts [T2-037/038; page footer]; product explainers (eFront Insight LP, Investment Cafe, Invest GP/LP, Portfolio Monitoring, Education Services) [T1-34/35; T2-078 to T2-082]; an "Ask eFront Copilot" video and a whole-portfolio promo with on-screen text only [T1-00 index]; and the Auto Commentary press release [page: Wealth Auto Commentary]. The Explore classroom video [T1-09] and the Fixed Income Performance Analyst walkthrough [T1-06] are on third-party channels.

## 3.7 Hiring and roles seen in the pulls

Job postings are summarised in the Studio briefing, with three cited: Senior Product Director, Aladdin Studio Developer (New York); Technical Writer, Aladdin Studio, Associate (New York); Studio Developer Operations, Vice President (Gurgaon) [briefing: Studio]. The product posting calls Studio Developer "Aladdin's extensibility layer", letting clients "build, adapt, and run custom workflows in a governed, enterprise-ready way", with goals to drive adoption "with BlackRock's own teams first, then clients", to "reduce off-platform development risk" and to embed governance in the developer experience; it names "AI-assisted developer tooling" [briefing: Studio]. The technical writer postings describe "Streaming APIs and webhook subscriptions, documented with AsyncAPI alongside OpenAPI; docs generated and versioned in CI/CD" [briefing: Studio]. The operations posting asks for "Python or Go plus Kubernetes/AKS" on an "Aladdin Container Platform: Kubernetes-based, Go control plane, operator pattern, AKS, Prometheus and Grafana"; data platform roles describe "scalable, governed, event-driven data ecosystems" with Kafka-style streaming [briefing: Studio]. The briefing also records that postings call Aladdin a "$1BN+ technology business" [briefing: Studio].

A Forward Deploy Engineer posting naming Model Context Protocol, a Technical Writer posting naming an "Aladdin Studio Developer Portal", an Aladdin Graph engineering posting citing "over 200 institutions", a Technical Solutions Architect building a citizen-developer community, and "free and premium capabilities" for Studio appear only in the candidate's revised draft; the fact-check lists these as unverifiable from the pulled material [fact-check A3].

The "citizen developer" vocabulary is BlackRock's own and predates the SDK. Khara, November 2023: "citizen developers are what are driving these organizations today" [T2-010]. Braunstein, March 2023: "this notion of the citizen developer. So, no matter which part of the organization you're in, you understand data, you understand data visualization, you understand how to do basic coding", and "we call that program broadly citizen development" [T2-022]. Woodhead: "inside of black rockck we have 3,000 citizen developers in business engineers that are interacting with all that in using Python or using R. I think that number has probably peaked" [T1-01]. Goel: "every organization has a growing number of citizen developers, or in fact, some people like to call them as embedded engineers now because they actually have a PhD degree in some cases" [T2-050]. A Big Little Questions speaker: "this new emerging class of citizen developers at an organization" [T1-25]. Garbien: "give technologists or the citizen developer the opportunity to work on really cool things" [T1-18]. On talent, Nair: "we have always tried to find talent at all levels that is fundamentally curious", and "I actually don't think it's going to

change the nature of the type of person that we're going to want" [T2-033]. Woodhead: "the grads that are coming in from college they have those skills built in" [T1-01].

## 3.8 Culture and operating statements

- "we are a user provider at the core. Ultimately, we eat our own cooking at BlackRock." Kunal Khara, MD and Global Head of Aladdin Product [T2-061].
- "BlackRock, the house, is using the same software, right? We have skin in the game in terms of the software we deploy." Sue Zheng, Tech Fellow [T2-027].
- "We've always had this principle, in building and operating Aladdin of stability being our first feature. We actually described stability, resilience, scale, performance, security, as a feature." Lance Braunstein [T2-037, May 2024].
- "I view the primary mission of the platform's organization as being number one, let's make sure Aladdin is up and running for our clients every single day." Afwa Kandawire, Global Head of Aladdin Platforms [T2-027].
- "What's important is the empowerment, making sure that with power comes responsibility. So, there's a control framework around that." Lance Braunstein [T2-022, Mar 2023].
- Governance for citizen code: "they have to have certain governance around checking this code into GitHub in the same governance level that we check code from our software engineer as well as they need to have survivorship so that if they leave the organization, we can make sure that the code still runs after they have left." Diwakar Goel [T2-050].
- "speed to market doesn't necessarily imply better data. You still have to have the data governance." Richard Newman, General Manager, Aladdin Studio [T1-11].
- "getting to trust is going to be predominantly about transparency in the short term. So, making sure that we've got human in the loop on the vast majority of operations, particularly in regulated industries, is going to be really, really critical." Adrian Crockett, Microsoft, on BlackRock's podcast; Garson's reply: "Can you show me how you got that answer? Can you cite me back to the original source?" [T1-12].
- "we are thinking deeply about what are the control frameworks and how does this operate more in the copilot paradigm than in the spit out the authoritative answer and that's my investment decision." Lance Braunstein [T2-022].
- "some of those core components that drive investment management as a process, human in the loop, control points, all those are then well integrated into a modernized, automated experience, very much navigated and driven by AI." Kunal Khara [T2-037].
- "We're not at a point of the agents making decisions. It's human in the loop." and "It's a continuous feedback loop as well in order to improve and to stay within the guardrails that are needed." Pavan Pemmaraju and Jess Fitzgibbon; the Whisper file does not label which of them says which [T2-028, Jul 2026].
- "There are guardrails that we can put in place ... We can't replace the judgment of the trader. That is the secret sauce." Ben Keeler, Partner, Citisoft, on BlackRock's podcast; Cameron Skinner's follow-up: "Where do you want the human in the loop? Where do you want them able to exercise their judgment" [T2-016, Jan 2026].
- "you're a lot in copilot, you must never provide investment advice ... build an evaluation pipeline and make sure that we are never going to give investment advice." Pedro Vicente Valdez, Principal AI Engineer [T1-

16]; the product page: "Will not give investment advice or respond to questions outside Aladdin platform boundaries" [page: Aladdin Copilot].

- Client co-development: "Our clients are the key driver behind how that product also evolves"; "the innovation does not come at the cost or should not ever come at the cost of anything being different or worse off in the way that they operate." Kunal Khara, Global Head of Aladdin Product [T1-30]. "I think our ability to do that without disrupting customers is kind of our superpower." Randy Fradin [T1-28]. "A good portion of the development that actually comes into the Aladdin system is on the back of the feedback that we receive directly from insurers." Georgette Anderson [T2-040].
- Efficiency: "Scale is often equated to do more with less. I think it's more do more with technology." Big Little Questions, asset managers episode [T1-24]. "Today we're seeing organizations trying to elevate talent. So move people from doing lower value add activities to much higher value add activities." Tim Garbien [T1-18]. "I think if you're thinking returns net of cost back to the client, you can return more when your costs are lower." Pia Schaefer [T2-011].
- "Never done" framing: "one thing that I can say with great confidence is never done is technology ... relevance is a point in time measurement." Sudhir Nair [T2-033]. "Never done, that's right." Lance Braunstein, agreeing with Jason Wild's "better, better, never done getting better" [T2-022]. "technology it wasn't furniture ... it was a lever, it was a catalyst that could actually transform the way business is done." Sudhir Nair [T2-033].
- "I just use that as another proof point of how some of the thinking of Aladdin was well ahead of its time because we've had this notion of an application framework embedded in the platform for quite a while." Afwa Kandawire [T2-027].

## 3.9 What the pulls do not show

The pulled transcripts and pages contain no technology-services revenue, margin, ACV or growth figure spoken by a BlackRock person; no count of Aladdin clients beyond "hundreds of clients globally across 70 countries" [T1-16] and the insurance figures in 3.1; no pricing, fee schedule or contract value for Aladdin, Studio, ADC or Copilot; no deal values for eFront, Preqin, HPS or GIP; no named wins or losses against competitors, and no comparison with Charles River, SimCorp, Enfusion, FactSet or any other vendor by a BlackRock speaker (the only competitor names in the corpus come from webinar panellists describing their own stacks). The AWS announcement, the Preqin-into-eFront-Insight date and the job-posting detail rest on the briefings, which drew on BlackRock pages outside the transcript set, and the Forward Deploy Engineer, Developer Portal and "over 200 institutions" items rest only on the candidate's draft. The NVIDIA GTC session and the Microsoft industry blog of September 2024 were not retrieved. These are outside the pulled material.